A

Project Report On

Society Management System

As a Partial Requirement for the Degree of BACHELOR

OF COMPUTER APPLICATION [B.C.A]

Submitted to



C.B. PATEL COMPUTER COLLEGE &

J.N.M. PATEL SCIENCE COLLEGE,

BHARTHANA, VESU, SURAT

Affiliated to

VEER NARMAD SOUTH GUJARAT UNIVERSITY, SUART

ACADAMIC YEAR: 2025-2026

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Introduction

- The Luxury Residential Society Management System (LRGMS) is a comprehensive full-stack web application designed to efficiently manage all aspects of a modern residential society. As residential communities grow, traditional manual management methods for tasks such as visitor entry, maintenance billing, and internal communication become inefficient, time-consuming, and prone to errors.

- This system provides a centralized digital platform that connects residents, society administrators, security personnel, and visitors, streamlining day-to-day operations and improving security, transparency, and convenience.

- The project is structured into three main components:

    1. **Main Frontend (Dashboard)** – Used by administrators, residents, and security guards to manage society operations, view notices, track visitors, and monitor financial activities. Built with React 19, Vite, Tailwind CSS, and Ant Design, it incorporates modern UI/UX elements with Framer Motion animations and interactive charts using Recharts and Chart.js.
    2. **Visitor Portal** – A lightweight interface for visitors to submit entry requests, pre-register visits, and receive notifications. It is designed for simplicity and mobile-friendly access, using React, Tailwind CSS, Framer Motion, and Swiper for interactive carousels.
    3. **Backend API** – The core engine that handles data storage, business logic, authentication, and third-party integrations. Built with Node.js, Express, and MongoDB, it supports secure authentication with JWT, real-time communication using Socket.io, automated notifications via Twilio, WhatsApp, and Nodemailer, and payment processing with Razorpay.

The system also features real-time notifications, QR-based visitor verification, automated maintenance billing, online payment integration, and data reporting/exporting, making it a reliable, scalable, and modern solution for managing residential societies efficiently.

Technology Used For Development

- ✓ **HTML**

HTML stands for HyperText Markup Language. It is the standard markup language used to create web pages. 10 HTML is a combination of Hypertext and Markup language.

- ✓ **CSS**

Cascading Style Sheets (CSS) is a style sheet language used for specifying the presentation and styling of a document written in a markup language such as HTML or XML.

- ✓ **Java Script**

JavaScript is a lightweight, cross-platform, single-threaded, and interpreted compiled programming 11 language.

- ✓ **React JS**

React also known as React.js or ReactJS is a free and open-source front-end JavaScript library that aims to make building user interfaces based on components more "seamless".

- ✓ **Node Js**

Node.js is a cross-platform, open-source JavaScript runtime environment that can run on Windows, Linux, Unix, macOS, and more. Node.js runs on the V8 JavaScript engine, and executes JavaScript code outside a web browser.

- ✓ **Express JS**

Express.js was founded by TJ Holowaychuk. The first release, according to Express.js's GitHub repository, was on 22 May 2010. Version 0.12

- ✓ **Mongo DB**

MongoDB is a source-available, cross-platform, document-oriented database program. Classified as a NoSQL database product, MongoDB uses JSON-like documents with optional schemas. Released in February 2009 by 10gen.

Objectives

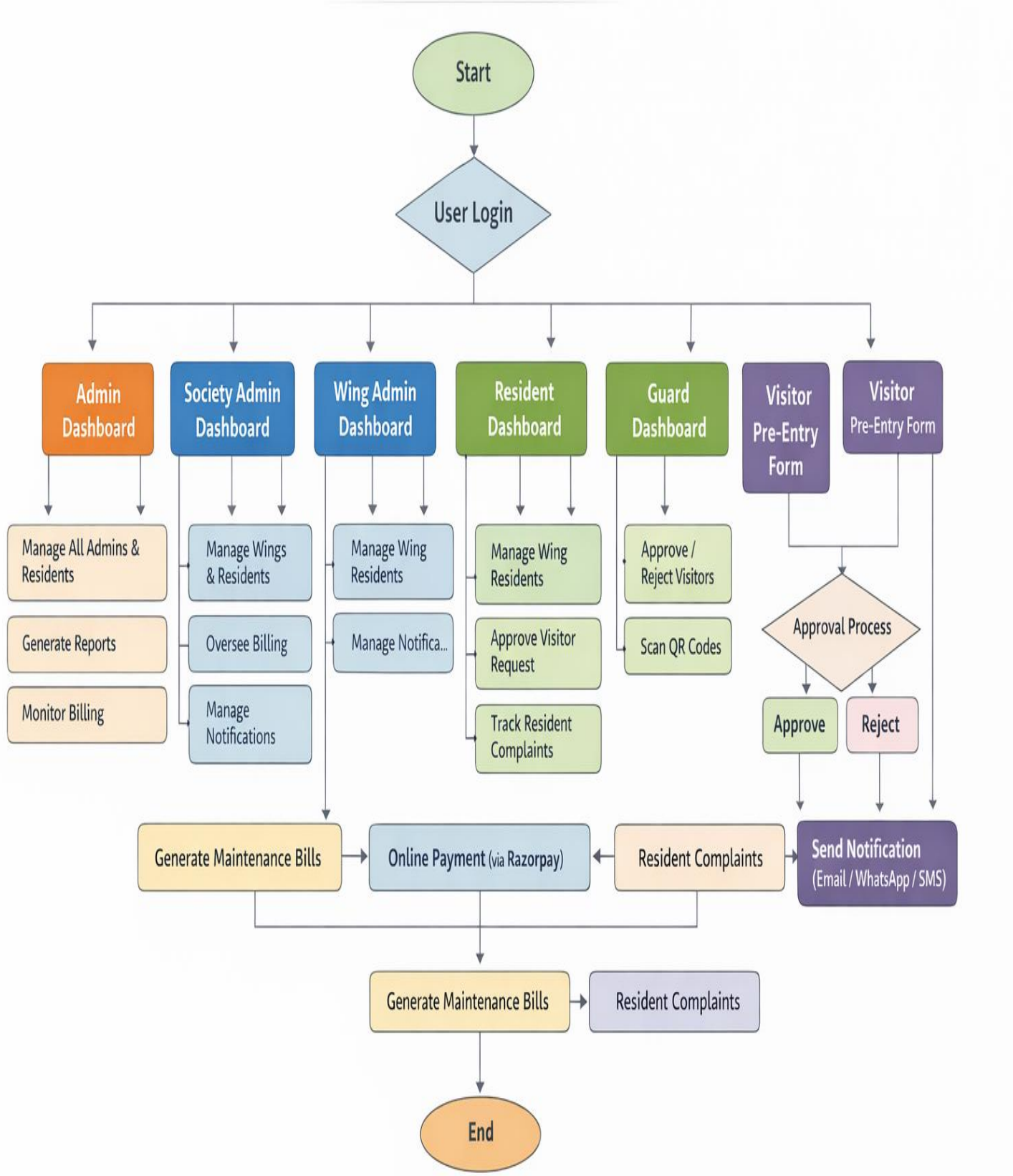
General Objectives :-

- To create an online platform for managing residential society operations.
- To reduce the dependency on manual, paper-based administrative processes.
- To provide a secure and reliable system for residents, visitors, and administrators.

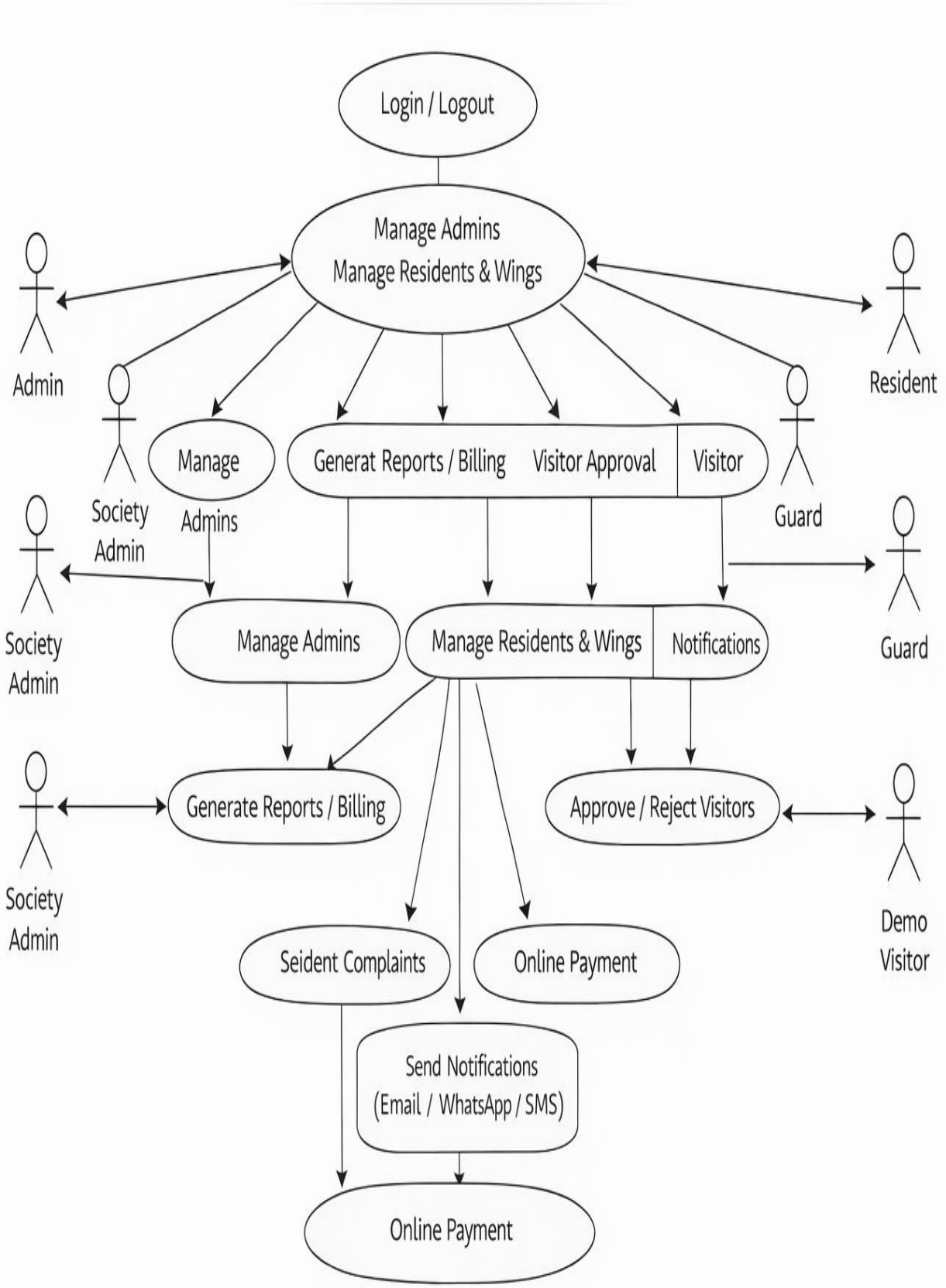
Specific Objectives :-

- To design a user-friendly interface using React.js and Tailwind CSS where residents and administrators can access society services easily.
- To implement secure authentication and role-based access for admins, residents, security guards, and visitors.
- To create a visitor management system that allows visitors to submit requests and receive approvals via WhatsApp, SMS, or email.
- To automate maintenance billing, payments, and financial reporting, including online payment integration via Razorpay.
- To provide a real-time notification system for alerts, announcements, visitor approvals, and emergencies.
- To design an admin dashboard that enables resident tracking, complaint management, billing oversight, and analytics.
- To maintain a MongoDB database that stores resident, visitor, billing, and society data efficiently.
- To implement QR-based visitor verification for enhanced security and entry tracking.

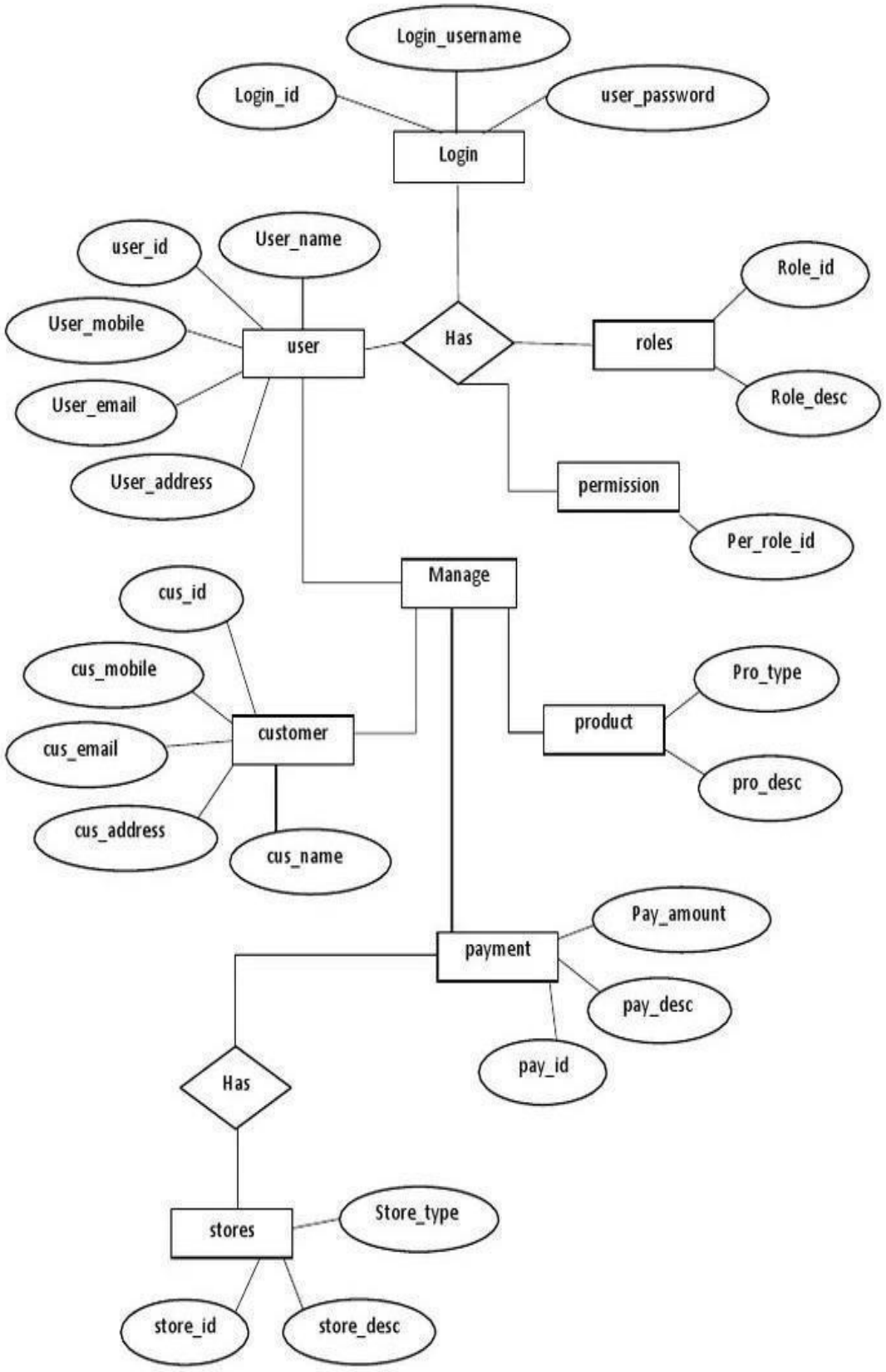
System Flow Chart



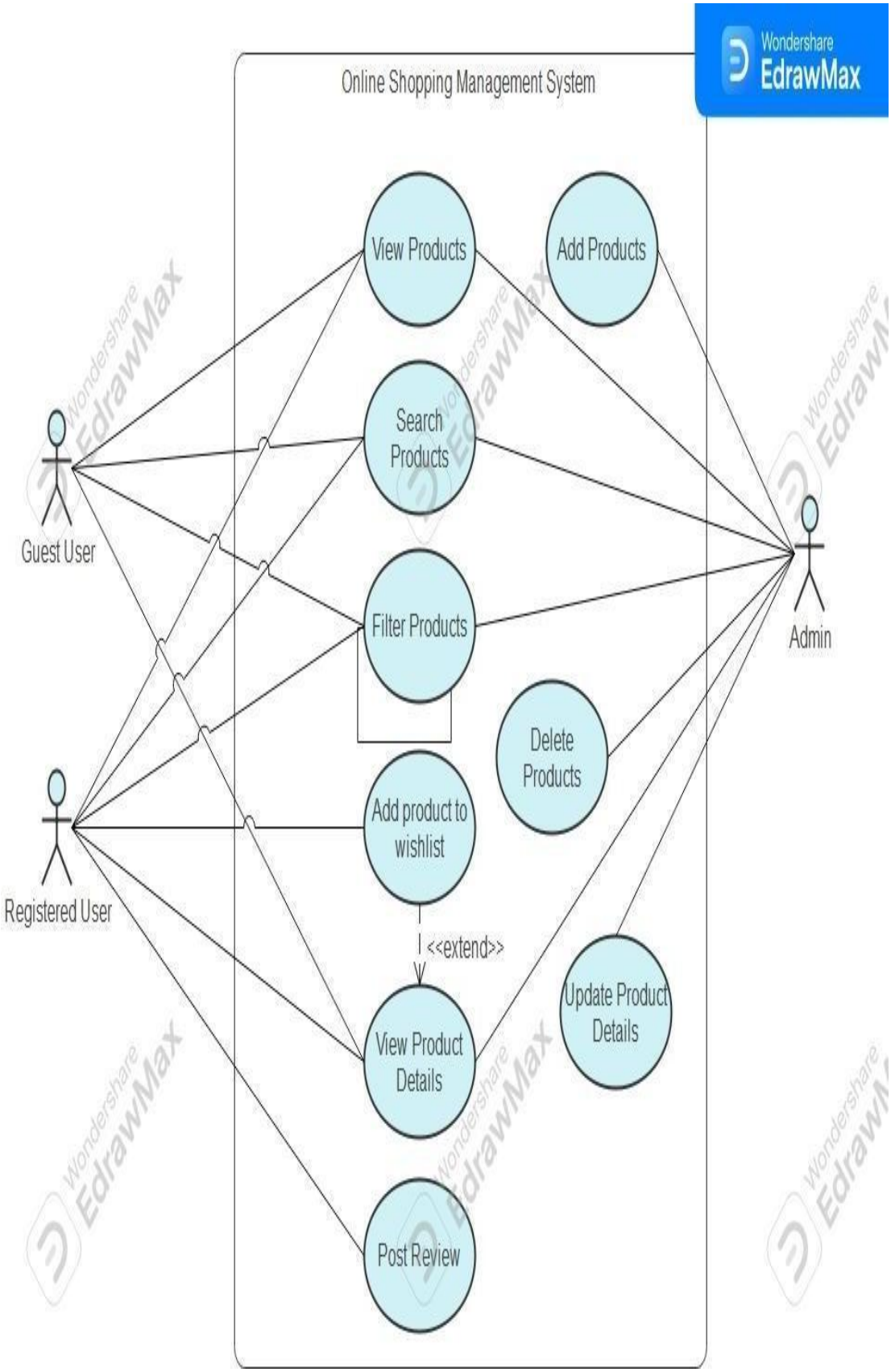
Use Case Diagram



ER Diagram


Login_username
Login_id
user_password
Login
user_id
User_name
Role_id
User_mobile
user
Has
roles
User_email
Role_desc
User_address
permission
Per_role_id
Manage
cus_id
cus_mobile
Pro_type
customer
product
cus_email
pro_desc
cus_address
cus_name
Pay_amount
payment
pay_desc
pay_id
Has
Store_type
stores
store_id
store_desc

Online Shopping Management System



Data Base Design

User :-

Field	**Type**	**Description**
_id	Objectid	Unique identifier
Name	String	Full name
Email	String(unique)	Login credential
Password	String	Hashed password
Role	Enum(customer,admin)	Access control
Timestamps	-	CreatedAt & UpdatedAt

Product:-

Field	Type	Description
_id	Object	
Name	String	
Description	String	
Price	Number	
Discount price	number	
Count in stock	Number	
Sku	String(unique)	Stock Keeping Unit
category	String	
Brand	String	
Sizes	[string]	Available sizes
Color	[string]	Available colors
Collection	string	Collection/series
Images	[{url, altText}]	Product images
Is featured	Boolean	Highlight in catalog
Is published	Boolean	Visibility control
rating	Number	Average rating
Num reviews	Number	Number of reviews
tags	[String]	Searchable tags
User	ObjectId (ref User)	Created by
Timestamps	-	CreatedAt & UpdatedAt

Cart:-

Field	Type	Description
_id	ObjectId	
Users	ObjectId (ref User)	Owner of cart
Guste id	String	For guest carts
Products	[{ productId, name, price, size, color, qty }]	Items in cart
Total price	Number	Total cart value
timestamps	-	CreatedAt & UpdatedAt

Order:-

Field	Type	Description
_id	ObjectId	
User	ObjectId (ref User)	Placed by
Order items	{ productId, name, image, price, size, color, qty }]	Items purchased
Shipping address	{ address, city, postalCode, country }	Delivery info
Pymentmethod	String	Payment type
Totalprice	Number	Final amount
Ispaid	Boolean	Payment status
Paidat	Date	Payment date
isdeileverd	Boolean	Delivery status
Deliveredat	Date	Delivery date
Paymentstatus	String	Gateway status
Status	Enum(processing, shipped, delivered, cancelled)	Order stage
timestamp	-	CreatedAt & UpdatedAt

Checkout:-

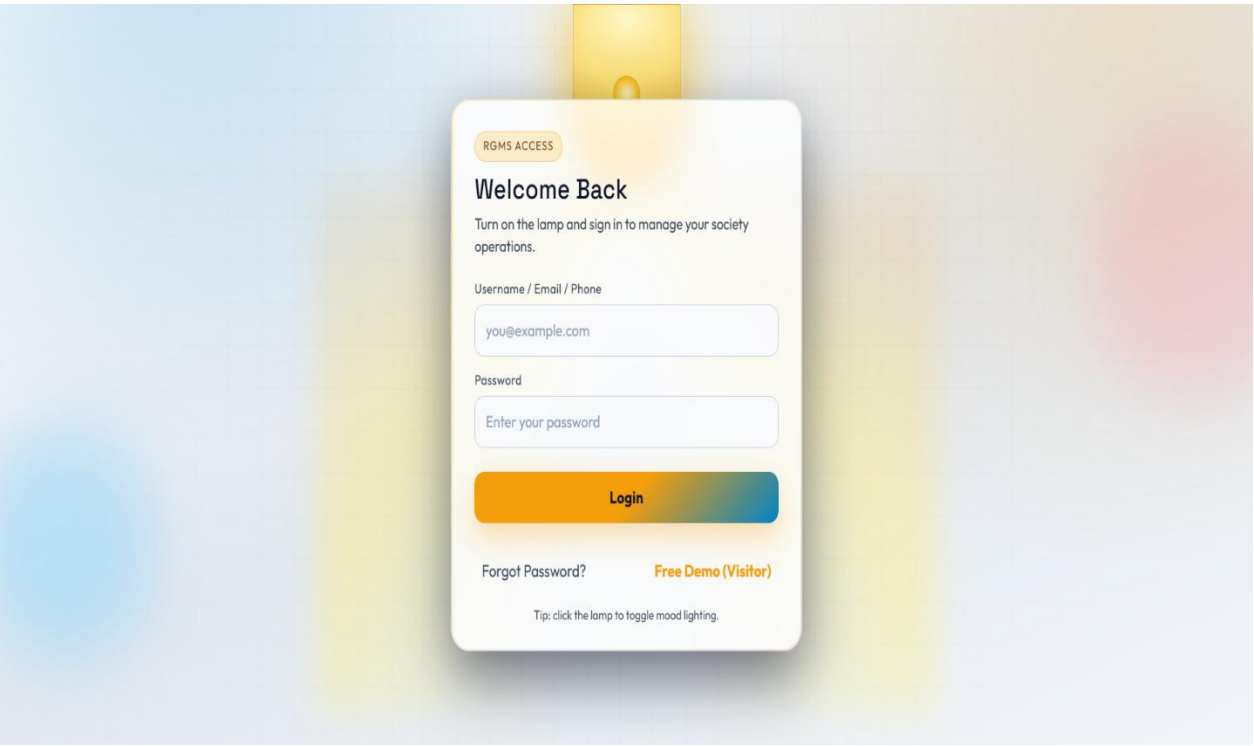
Field	Type	Description
_id	ObjectId	
User	ObjectId (ref User)	Linked customer
Cart	ObjectId (ref Cart)	Cart being checked out
Paymentintentid	String	From payment gateway
Status	Enum(pending, completed, failed, cancelled)	Transaction state
Amount	Number	Payment amount
Currency	String	Currency type
timestamps	-	CreatedAt & UpdatedAt

Subscriber:-

Field	Type	Description
_id	ObjectId	
Email	String (unique)	Subscriber email

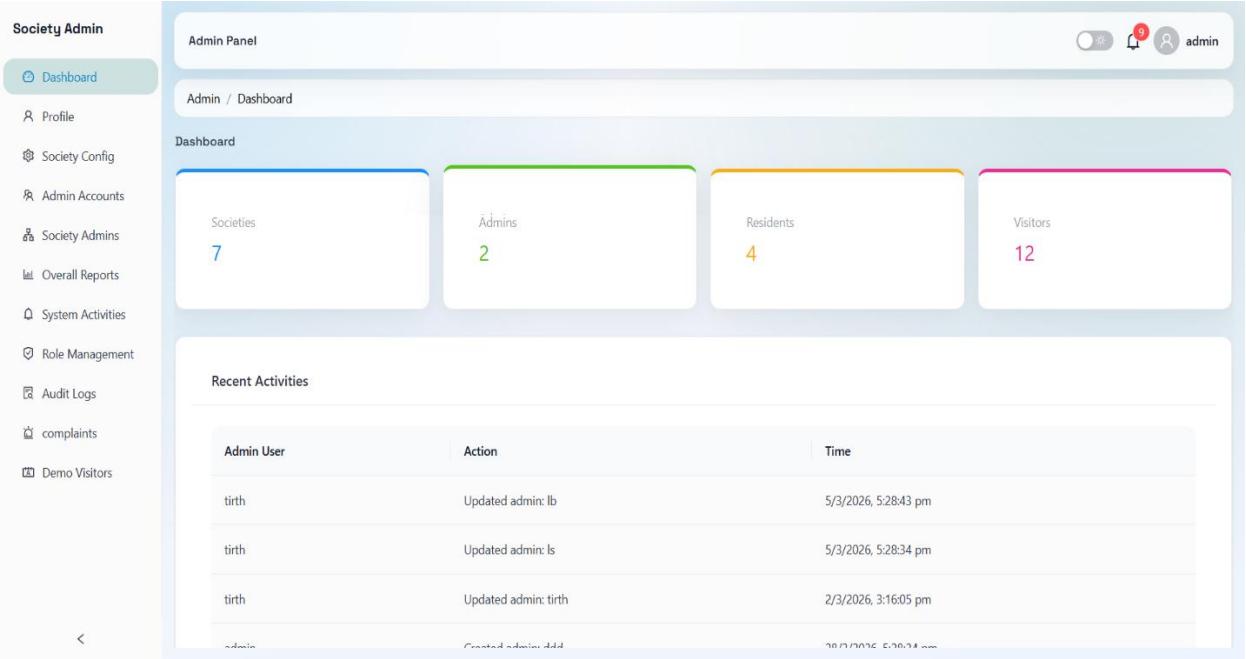
Screen Shots

Login Page:-

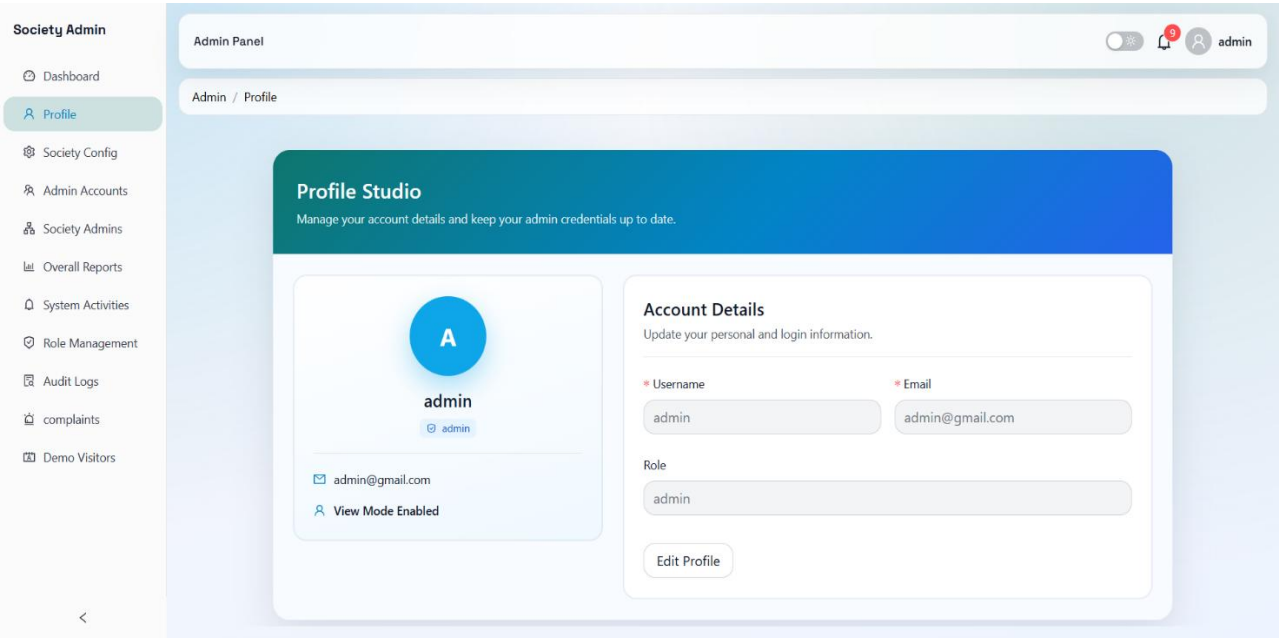


Admin :-

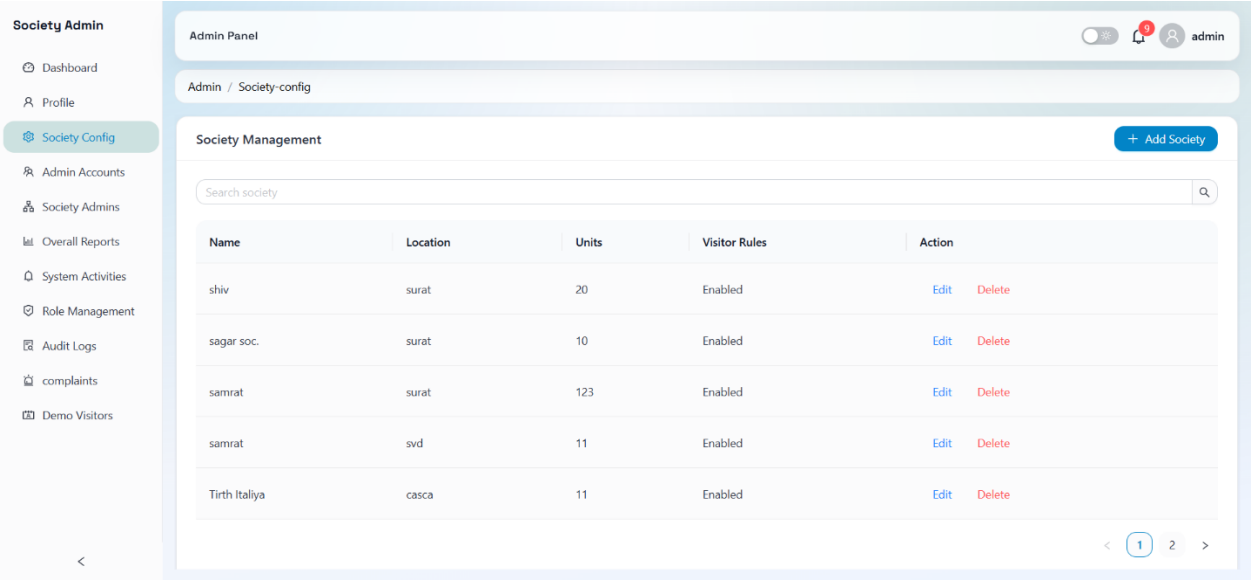
Dashboard :-



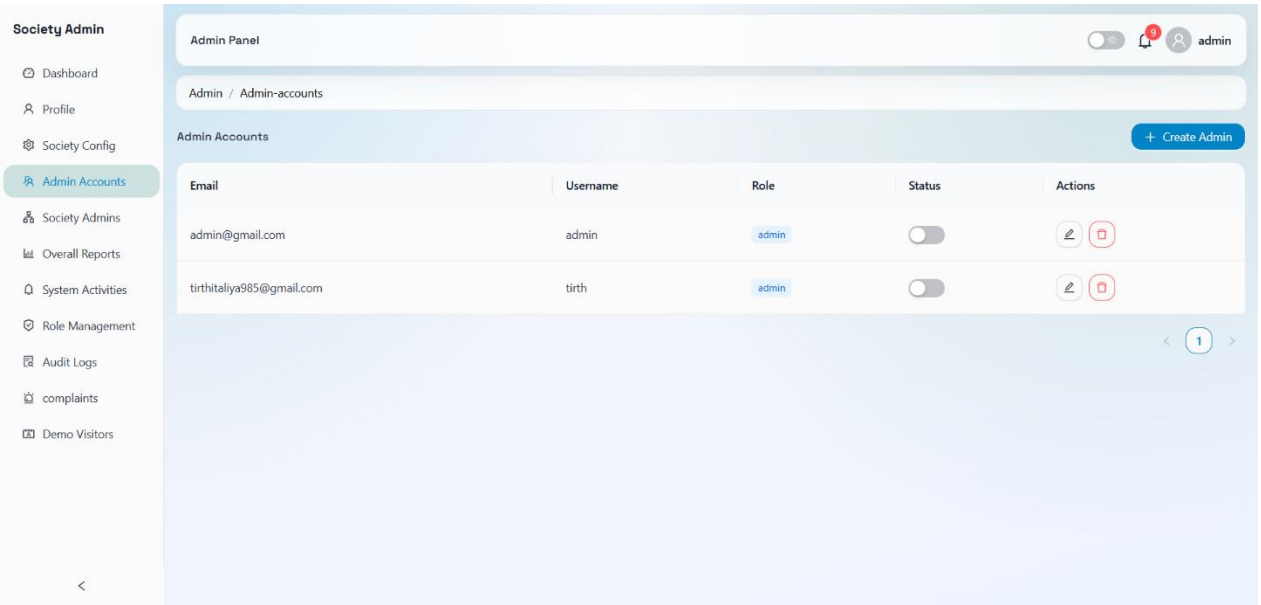
Profile :-



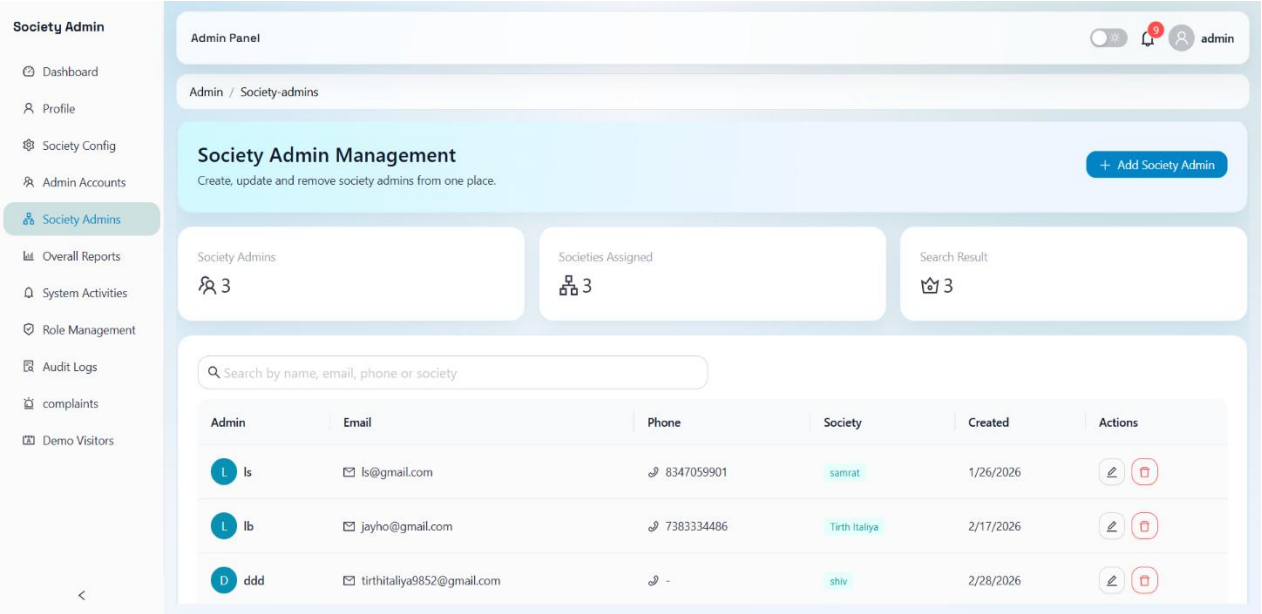
Society config:-



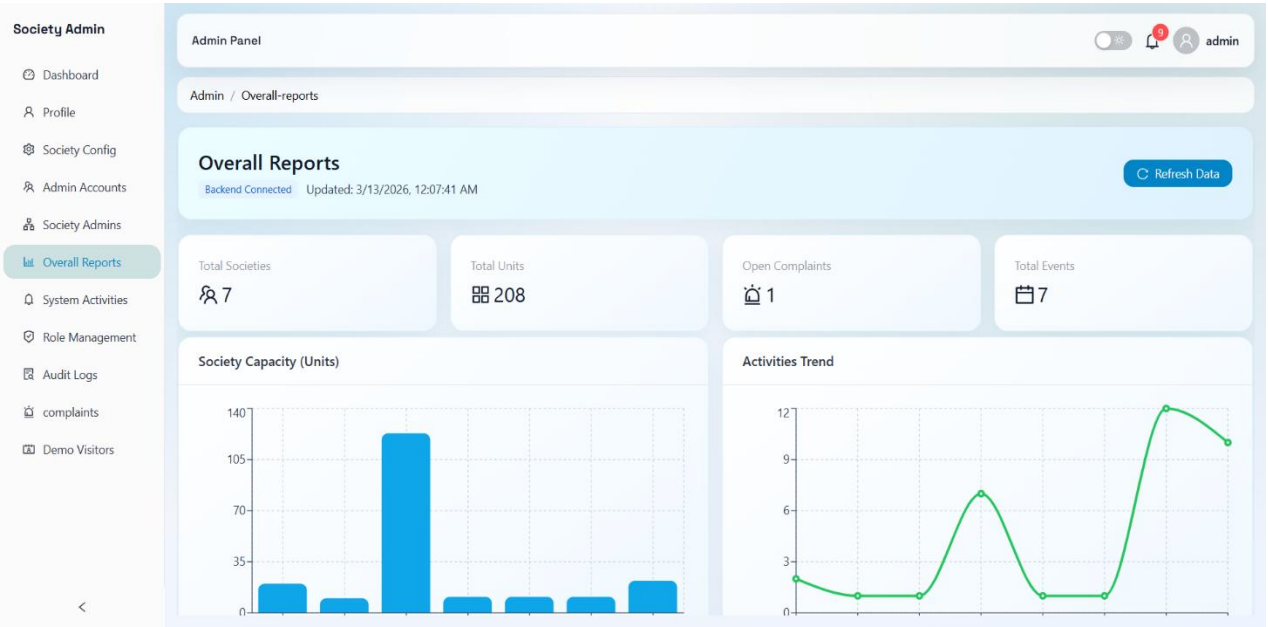
Admin Accounts:-



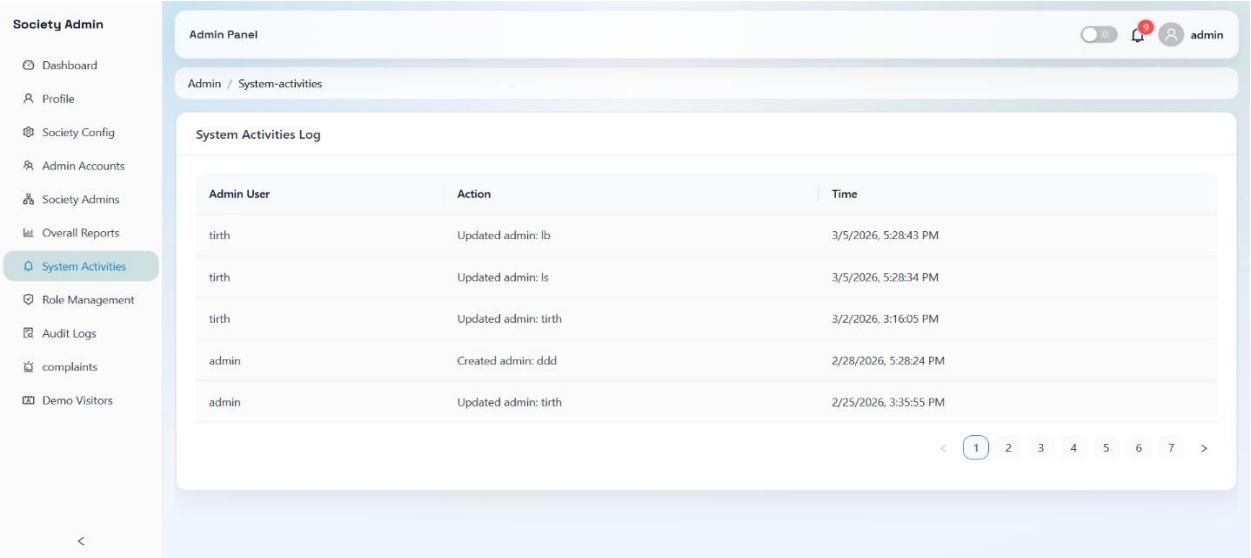
Society Admin:-



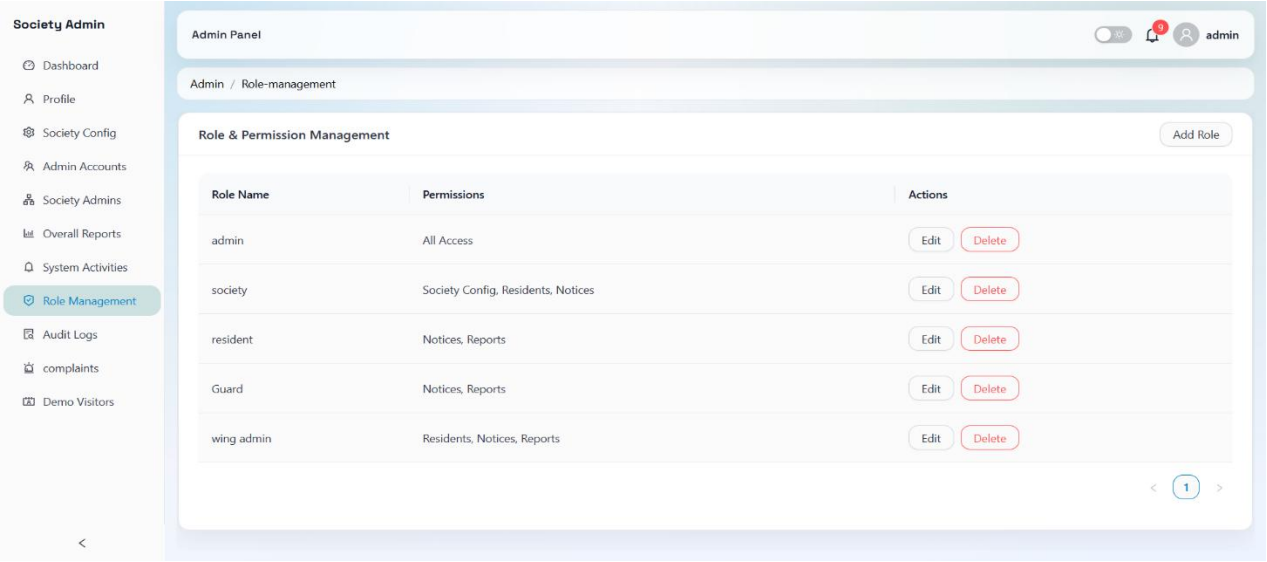
Overall Reports :-



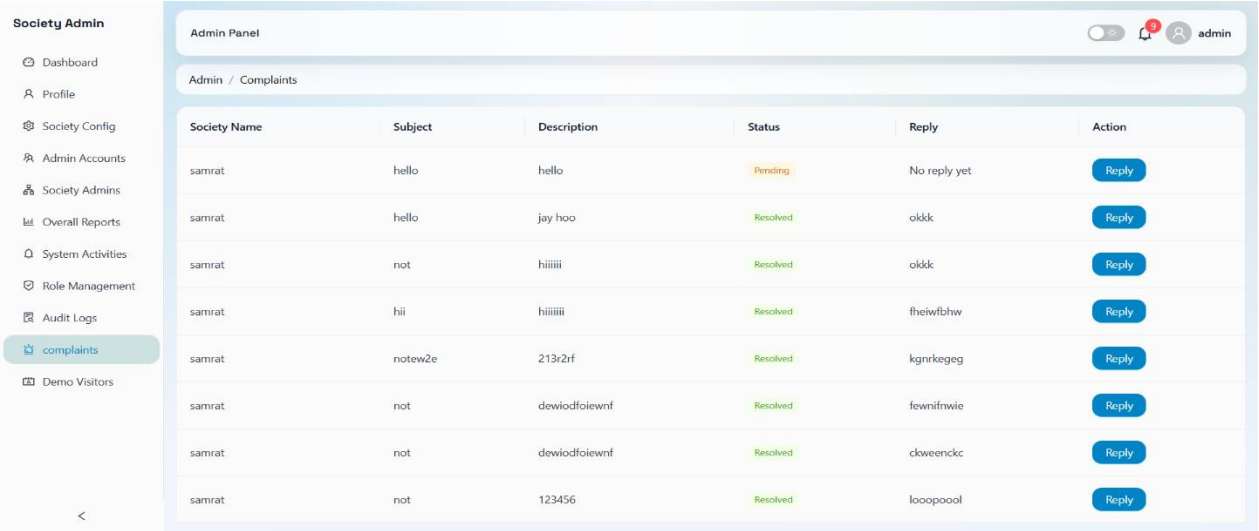
System Activities :-



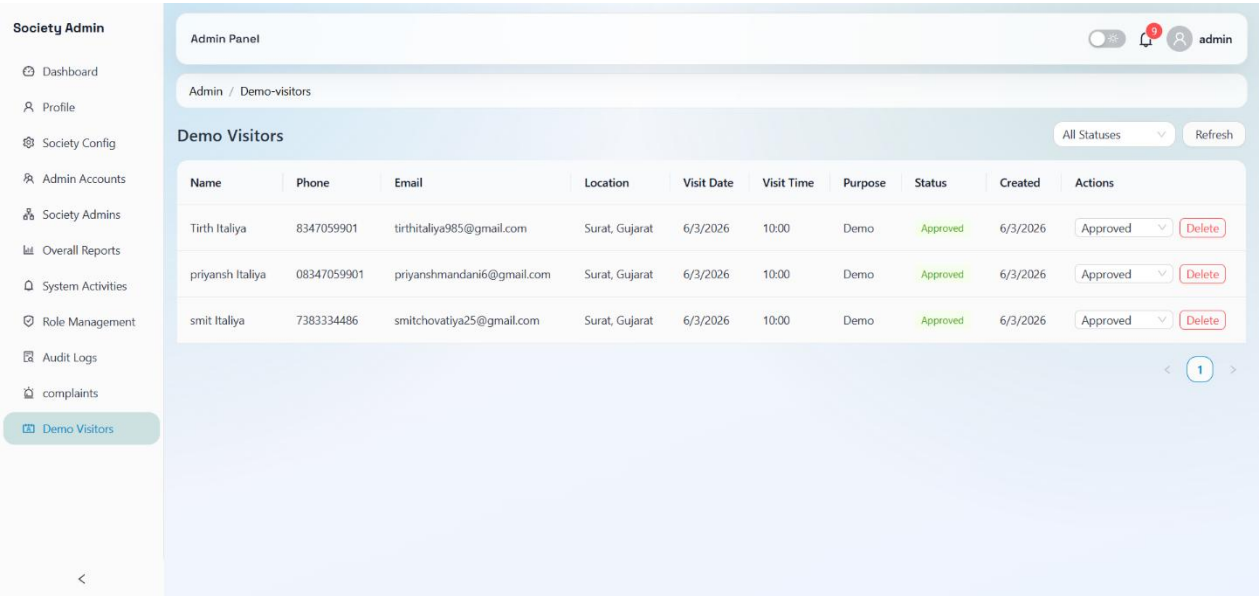
Role Management :-



Complains:-

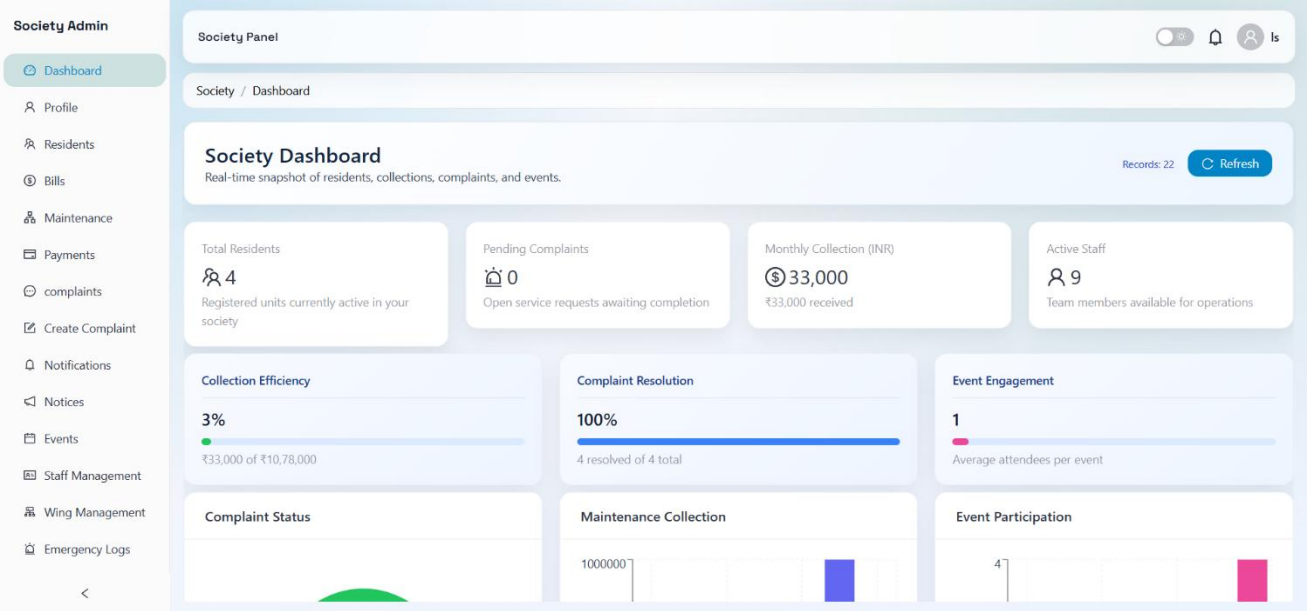


Demo Visitors :-

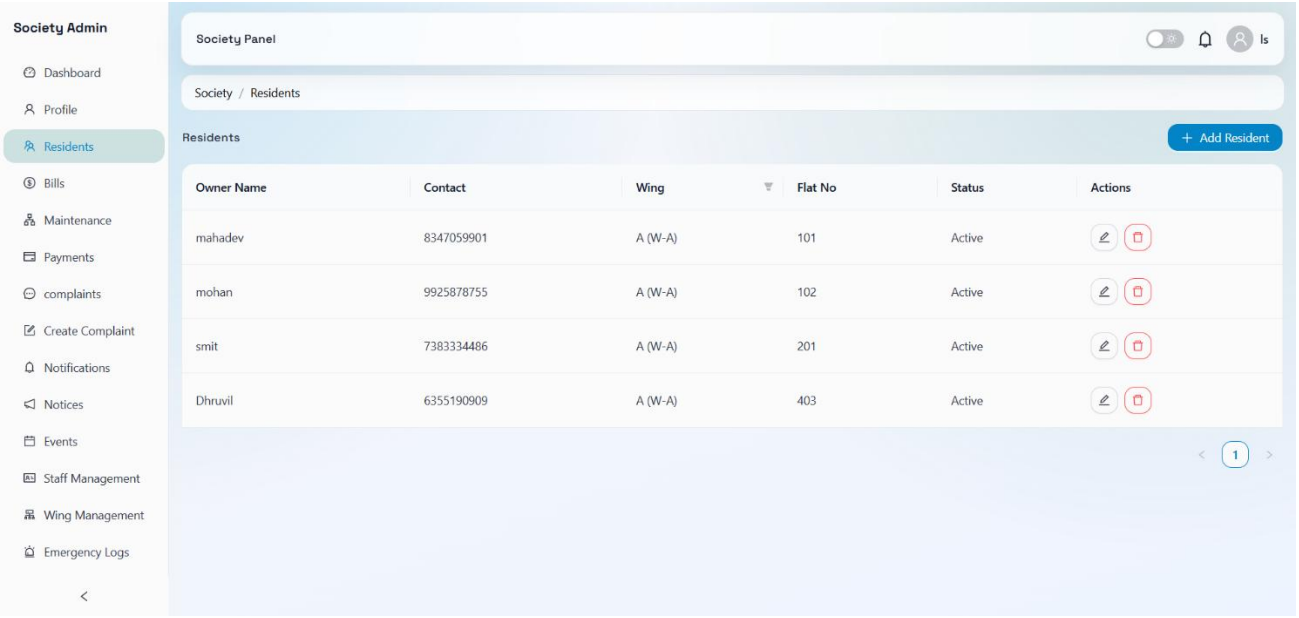


Society Admin :-

Dashboard:-

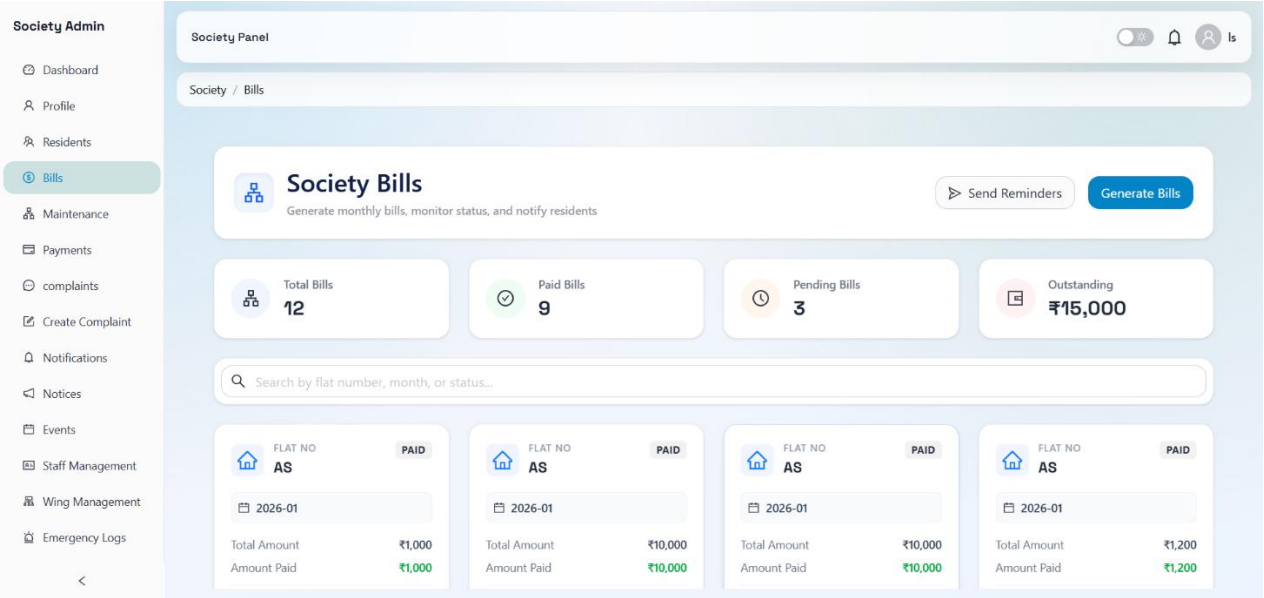


Residents :-

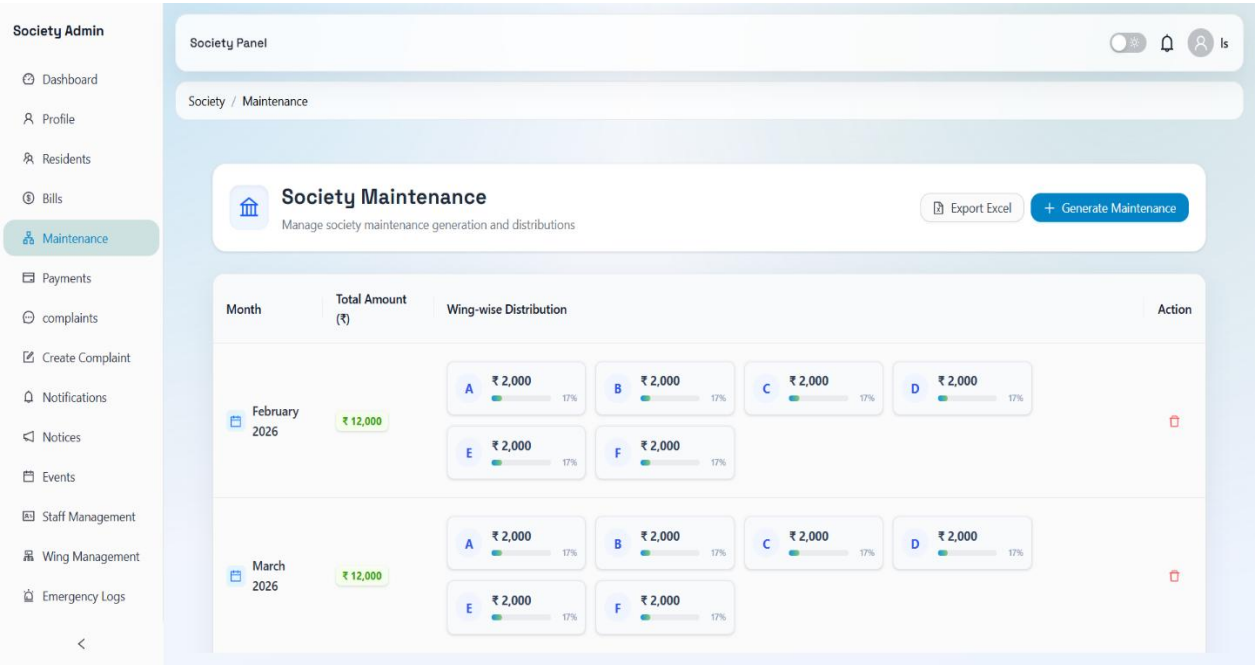


Style nest Clothing

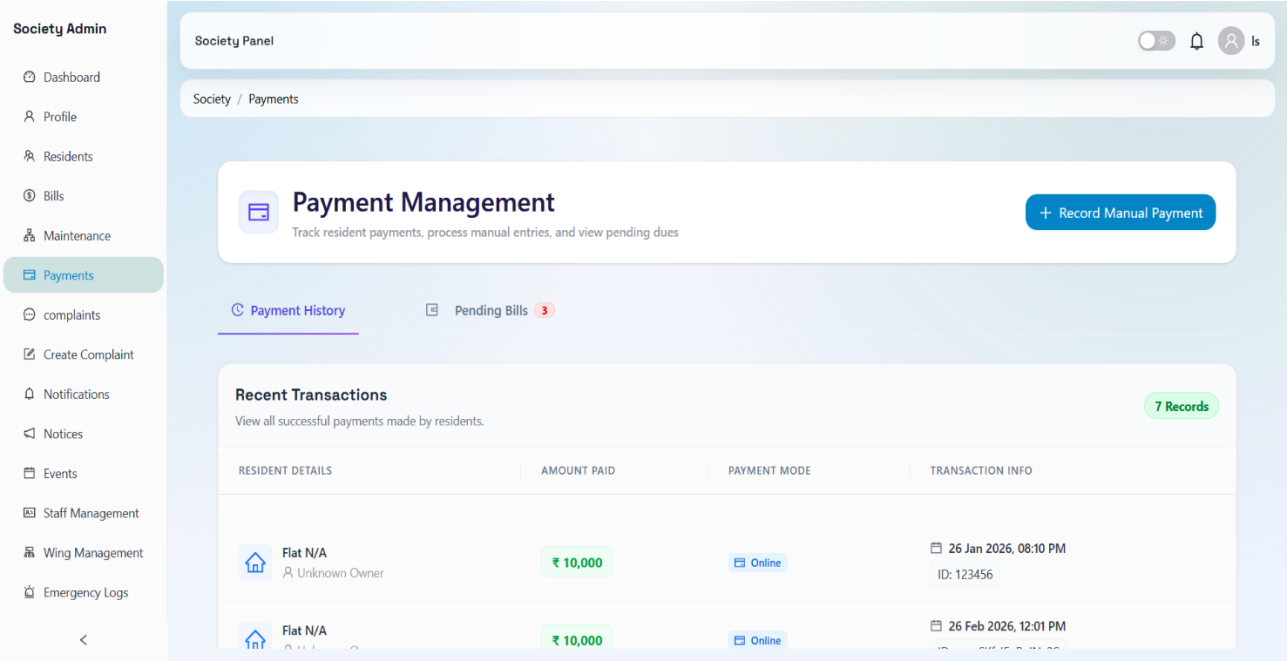
Bills :-



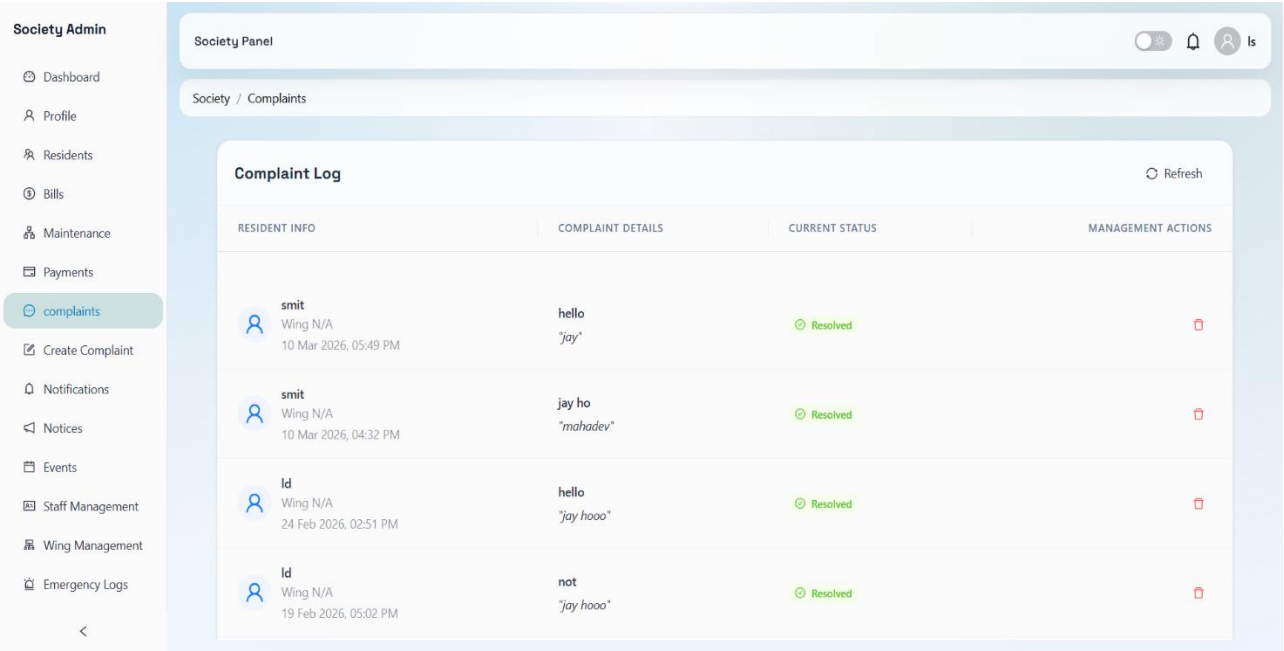
Maintenance:-



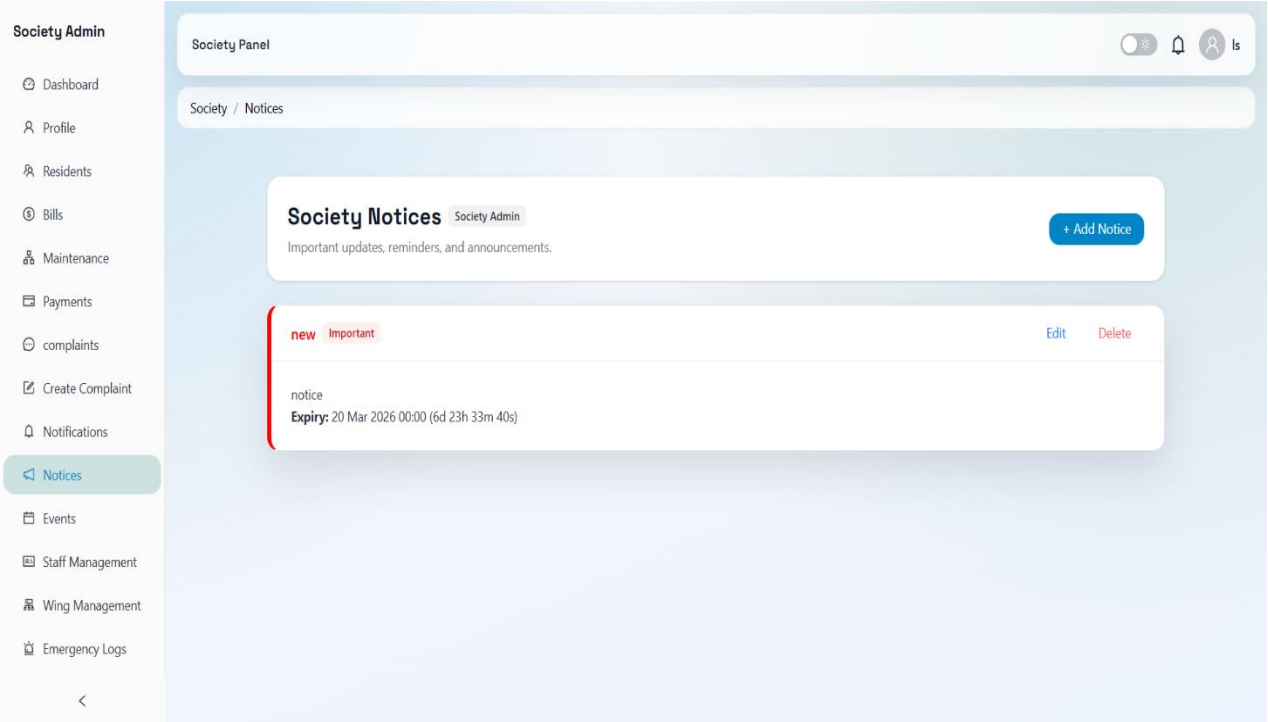
Payments :-



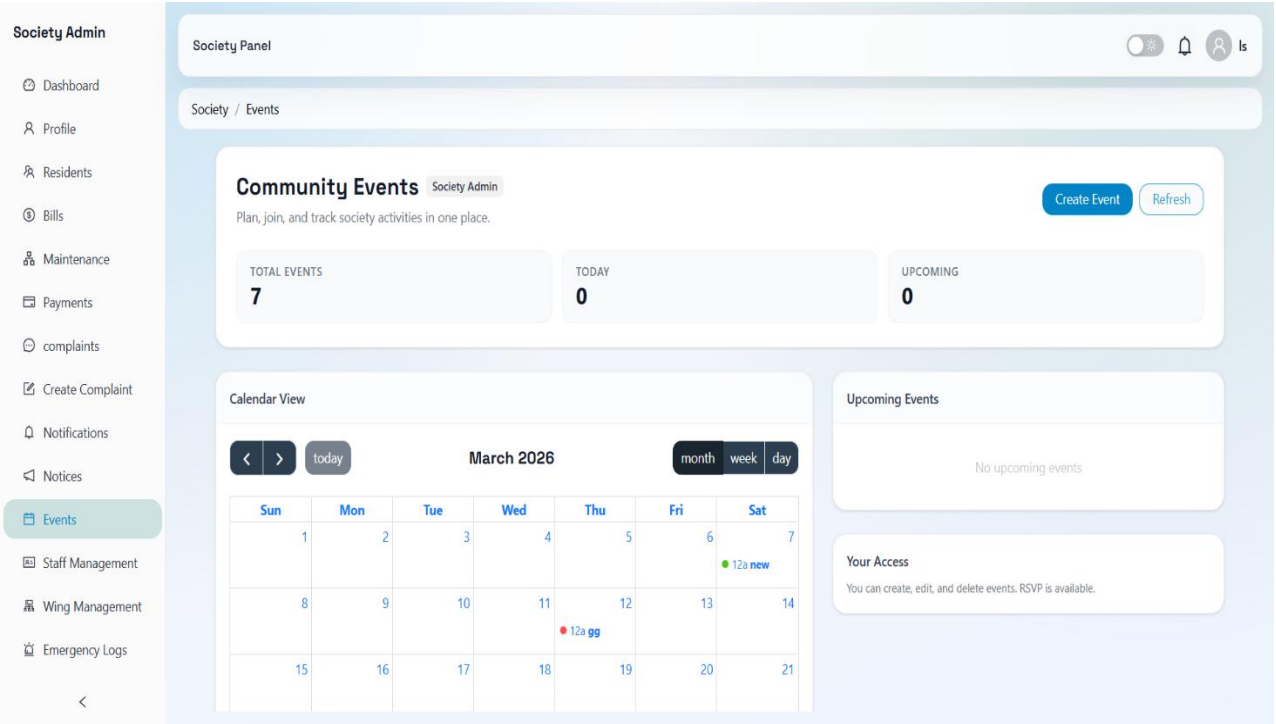
Complaints:-



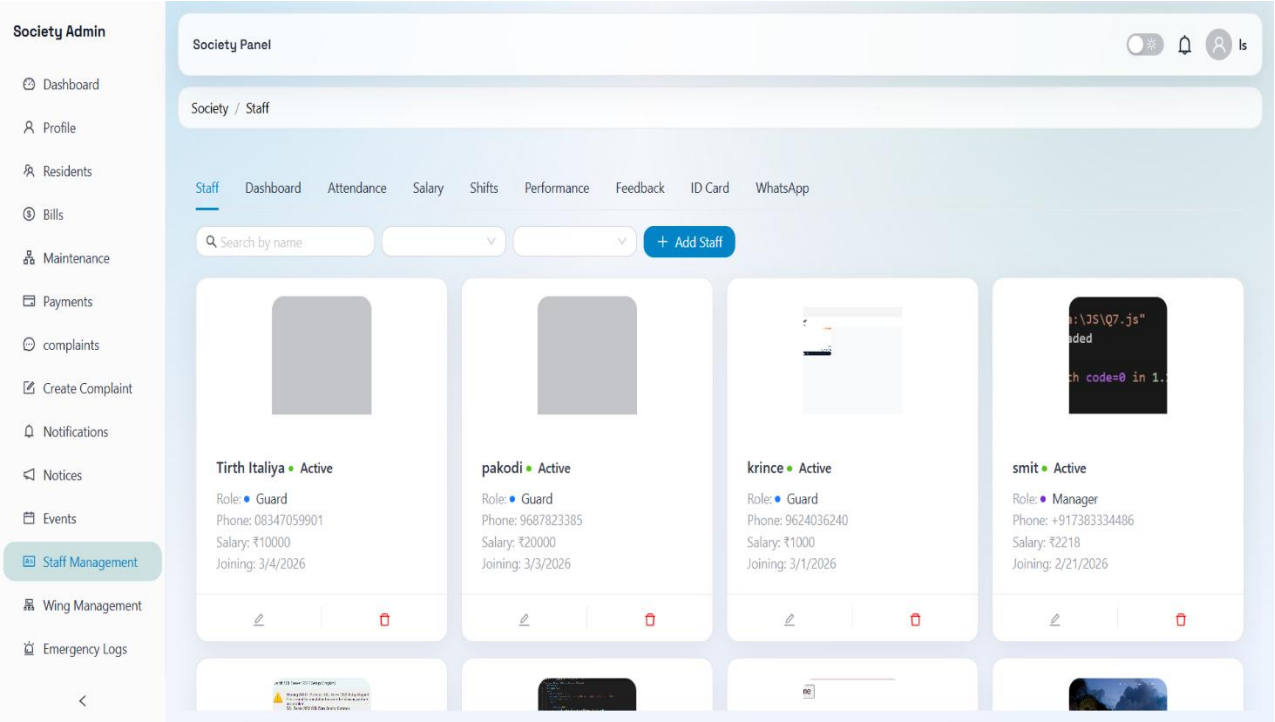
Notification :-



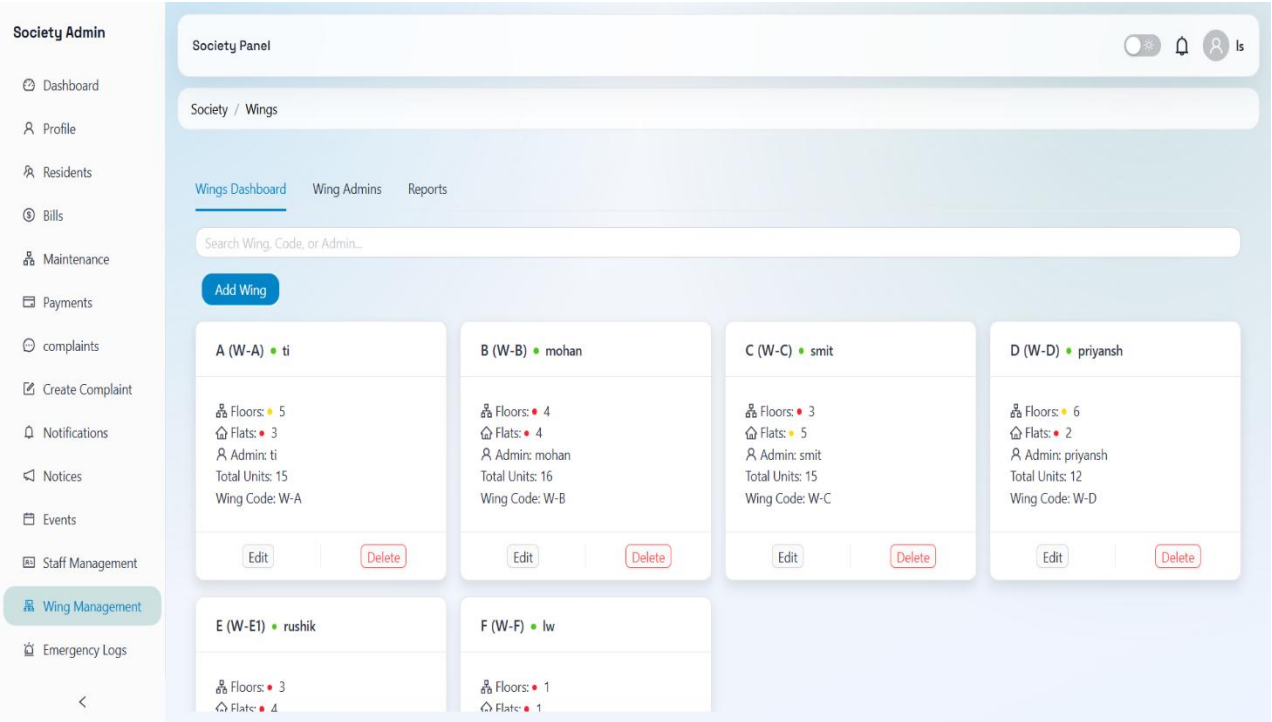
Events:-



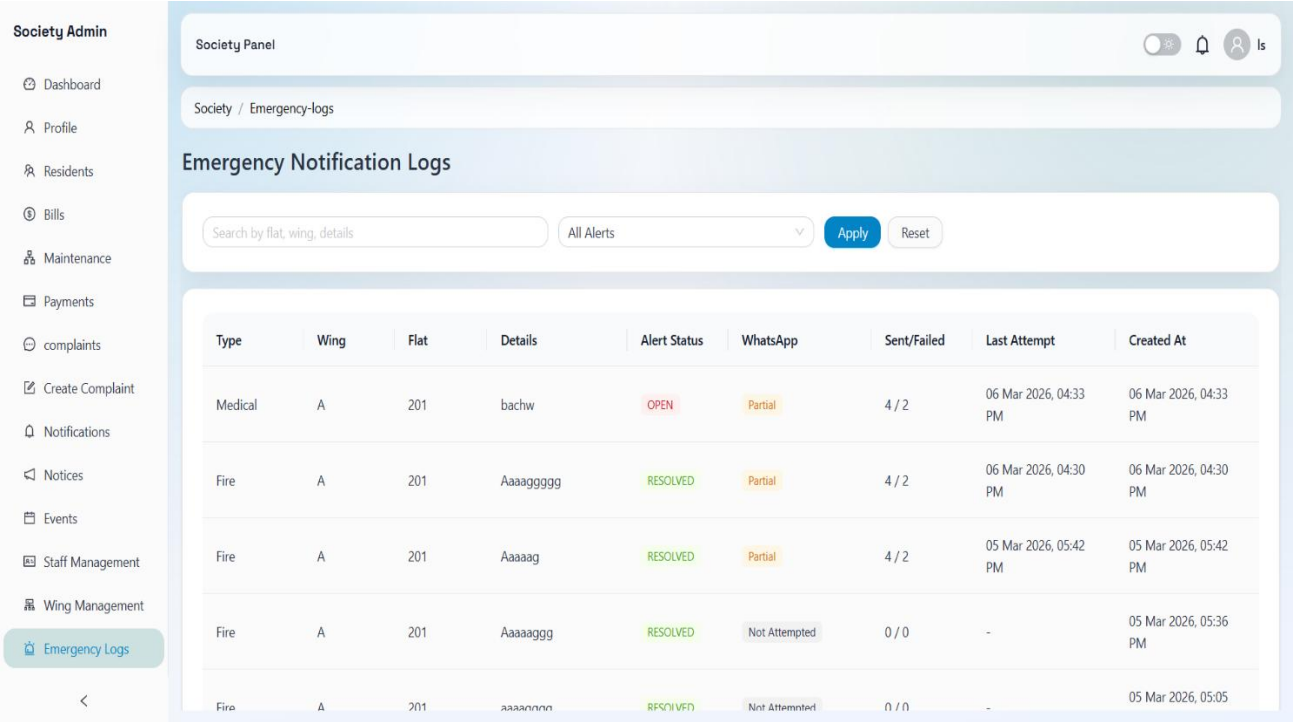
Staff :-


Society Admin
Dashboard
Profile
Residents
Bills
Maintenance
Payments
complaints
Create Complaint
Notifications
Notices
Events
Staff Management
Wing Management
Emergency Logs
Society Panel
Society / Staff
Staff
Dashboard
Attendance
Salary
Shifts
Performance
Feedback
ID Card
WhatsApp
Search by name
Add Staff
Tirth Italiya • Active
Role: • Guard
Phone: 08347059901
Salary: ₹10000
Joining: 3/4/2026
pakodi • Active
Role: • Guard
Phone: 9687823385
Salary: ₹20000
Joining: 3/3/2026
krince • Active
Role: • Guard
Phone: 9624036240
Salary: ₹1000
Joining: 3/1/2026
smit • Active
Role: • Manager
Phone: +917383334486
Salary: ₹2218
Joining: 2/21/2026

Wing :-

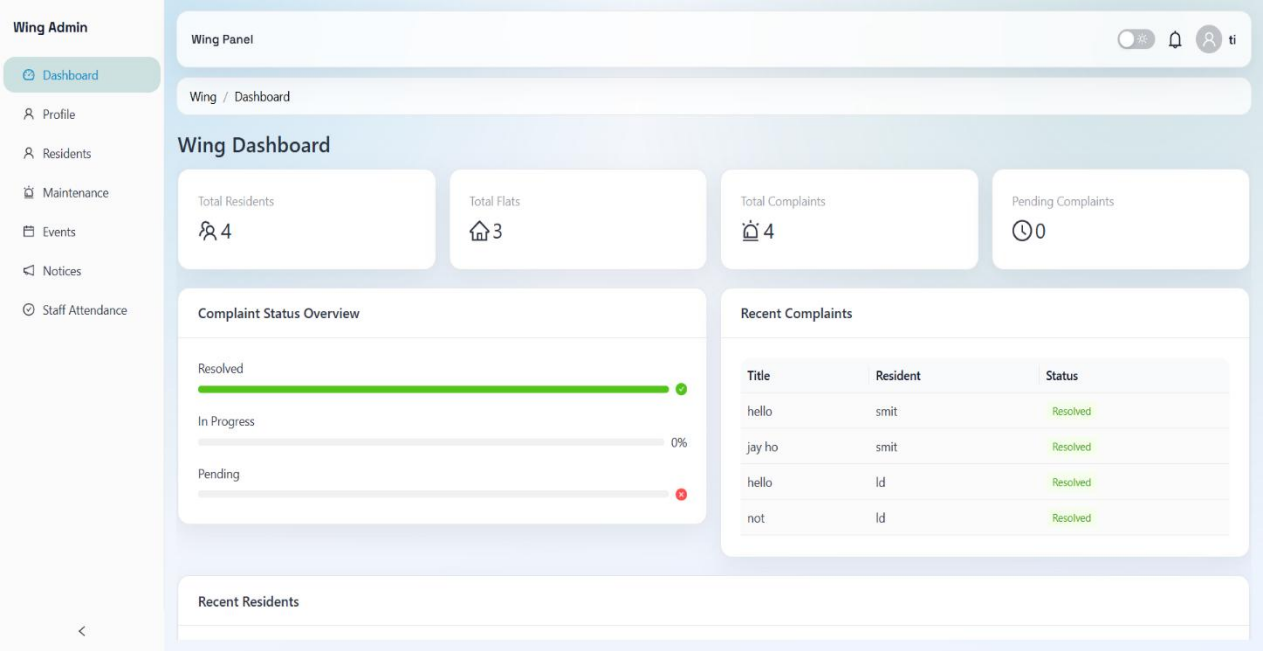

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Staff Management
Wing Management
Emergency Logs
Society Panel
Society / Wings
Wings Dashboard
Wing Admins
Reports
Search Wing, Code, or Admin...
Add Wing
A (W-A) • ti
Floors: • 5
Flats: • 3
Admin: ti
Total Units: 15
Wing Code: W-A
Edit
Delete
B (W-B) • mohan
Floors: • 4
Flats: • 4
Admin: mohan
Total Units: 16
Wing Code: W-B
Edit
Delete
C (W-C) • smit
Floors: • 3
Flats: • 5
Admin: smit
Total Units: 15
Wing Code: W-C
Edit
Delete
D (W-D) • priyansh
Floors: • 6
Flats: • 2
Admin: priyansh
Total Units: 12
Wing Code: W-D
Edit
Delete
E (W-E1) • rushik
Floors: • 3
F (W-F) • lw
Floors: • 1

Emergency :-

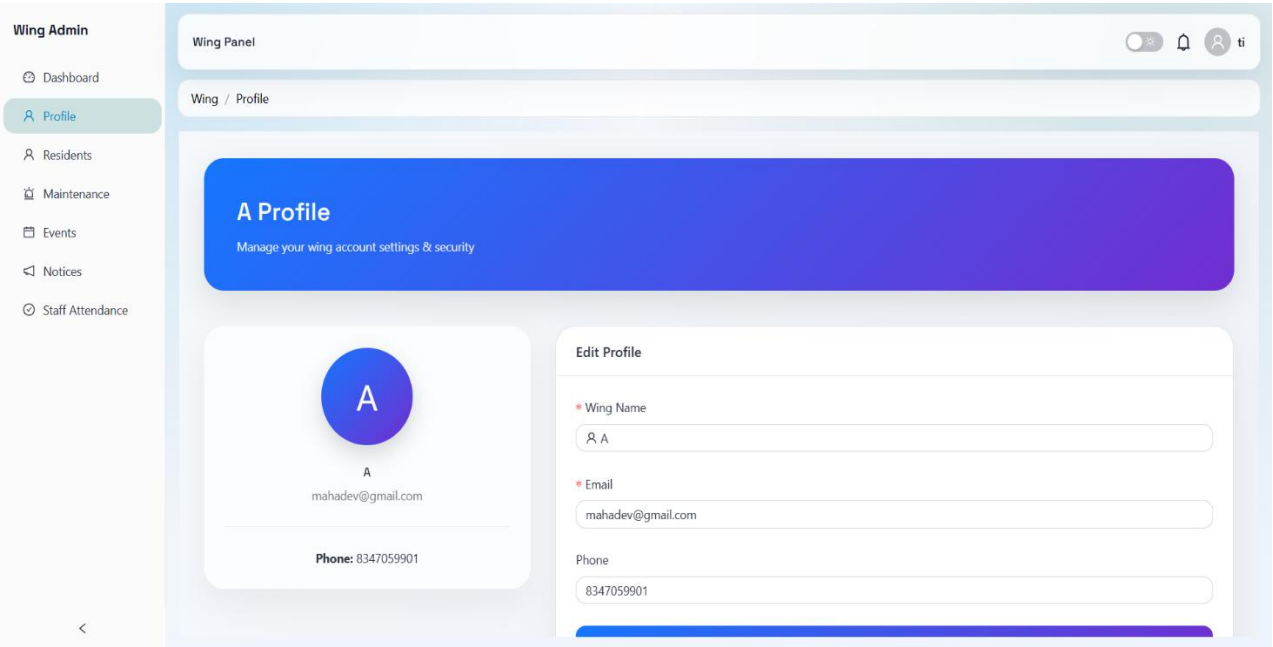

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Staff Management
Wing Management
Emergency Logs
Society Panel
Society / Emergency-logs
Emergency Notification Logs
Search by flat, wing, details
All Alerts
Apply
Reset
Type Wing Flat Details Alert Status WhatsApp Sent/Failed Last Attempt Created At
Medical A 201 bachw OPEN Partial 4 / 2 06 Mar 2026, 04:33 PM 06 Mar 2026, 04:33 PM
Fire A 201 Aaaaggggg RESOLVED Partial 4 / 2 06 Mar 2026, 04:30 PM 06 Mar 2026, 04:30 PM
Fire A 201 Aaaaag RESOLVED Partial 4 / 2 05 Mar 2026, 05:42 PM 05 Mar 2026, 05:42 PM
Fire A 201 Aaaaaggg RESOLVED Not Attempted 0 / 0 - 05 Mar 2026, 05:36 PM

Wing Admin :-

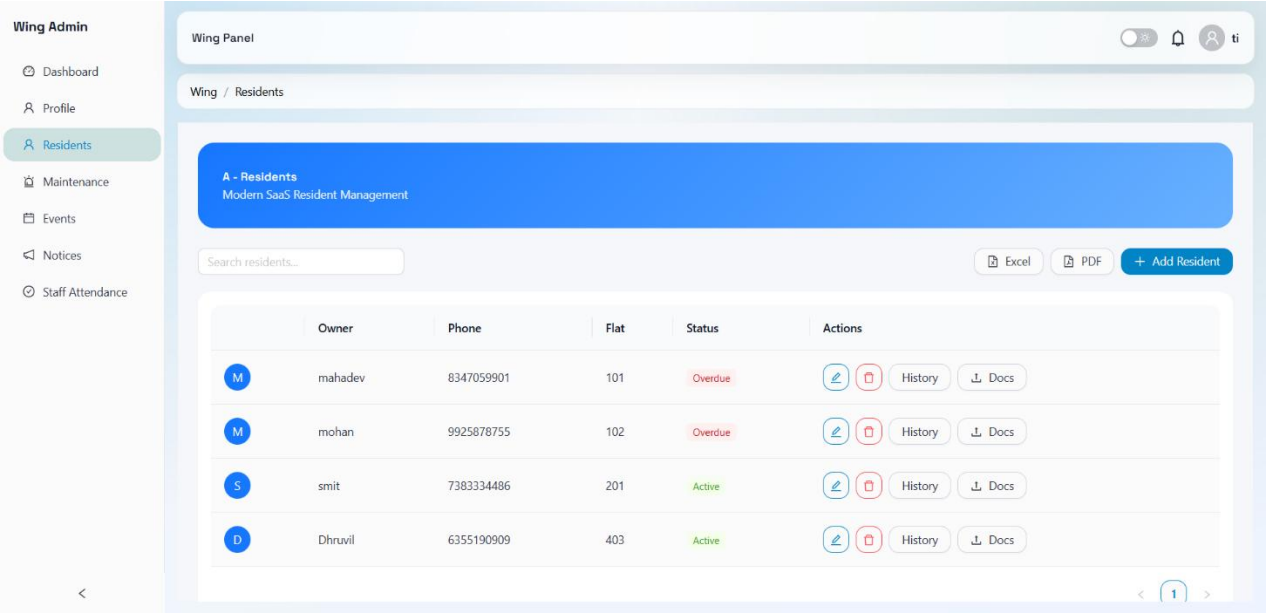
Dashboard :-



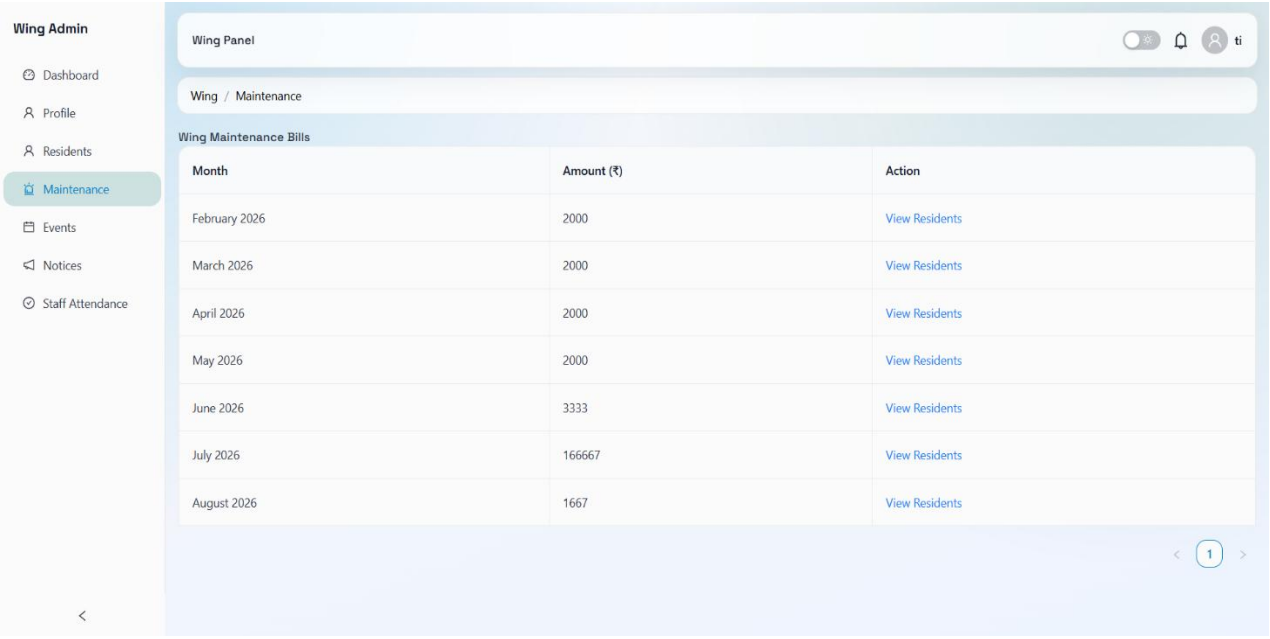
Profile :-



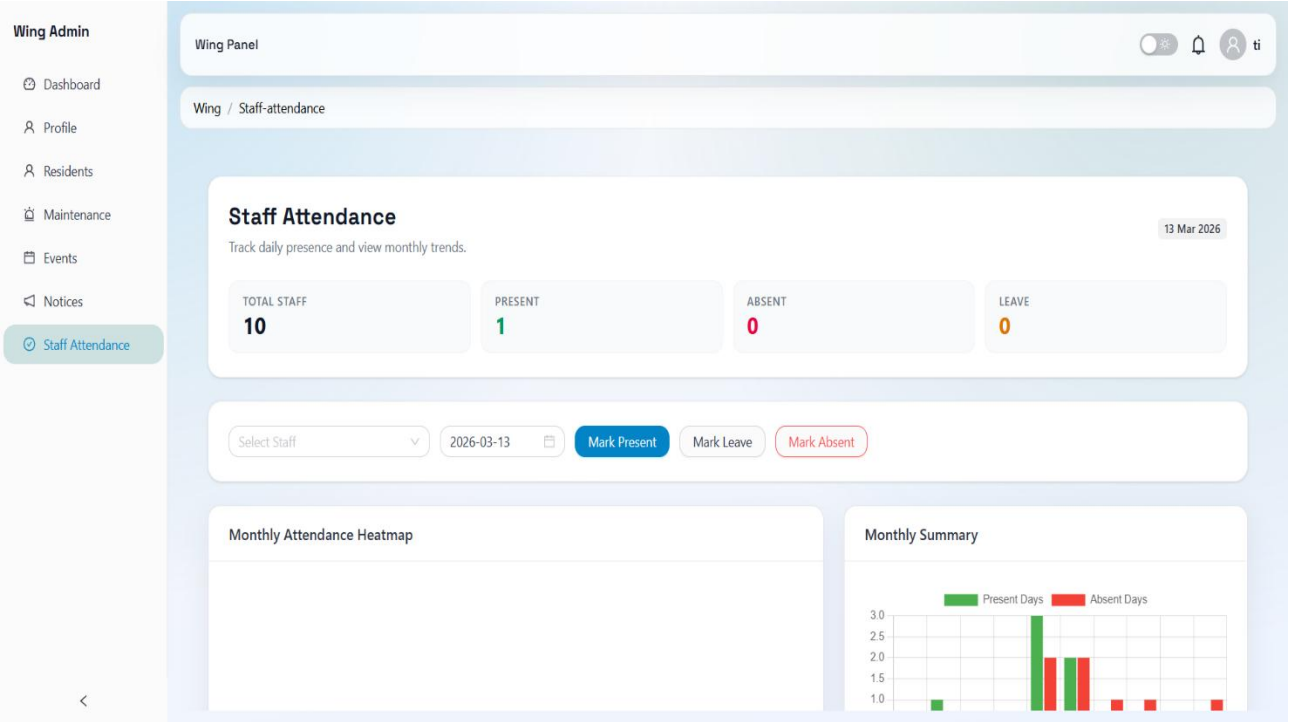
Residents :-



Maintenance :-

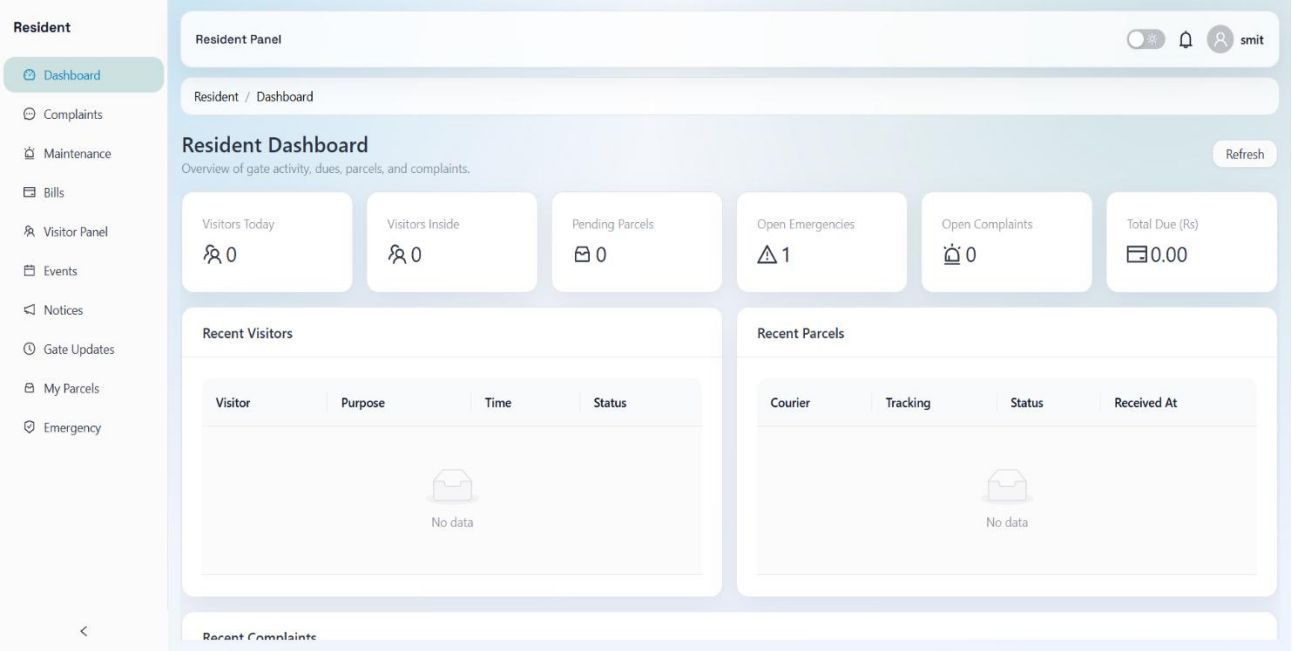


Staff Attendance :-

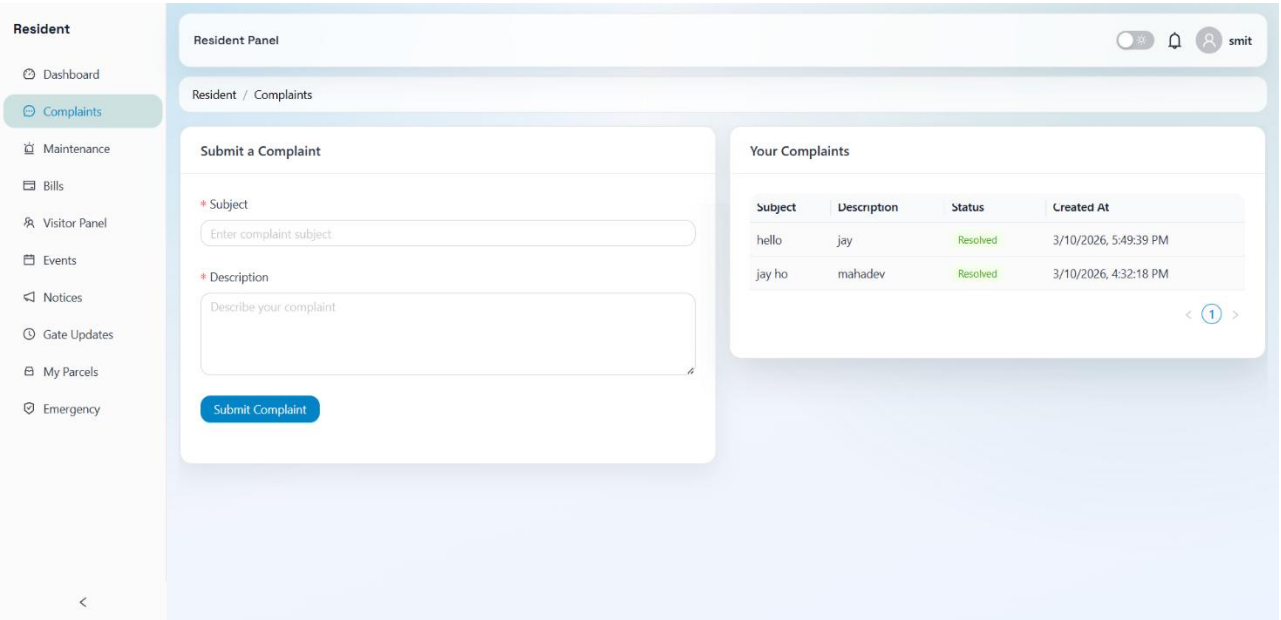


Resident :-

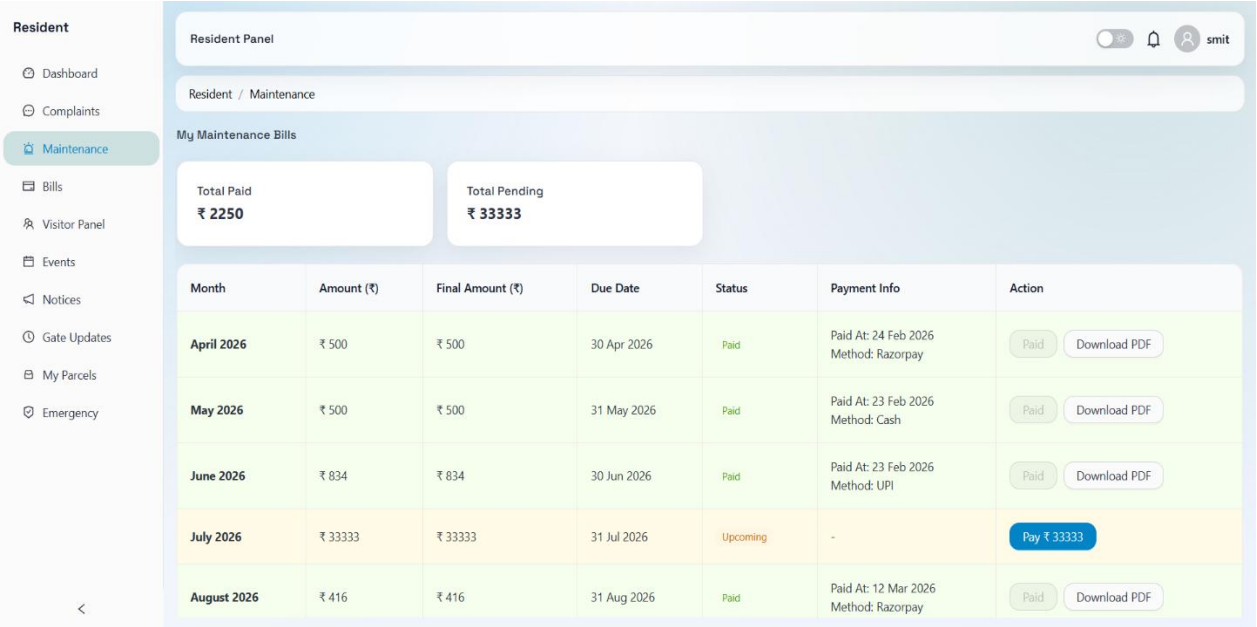
Dashboard :-



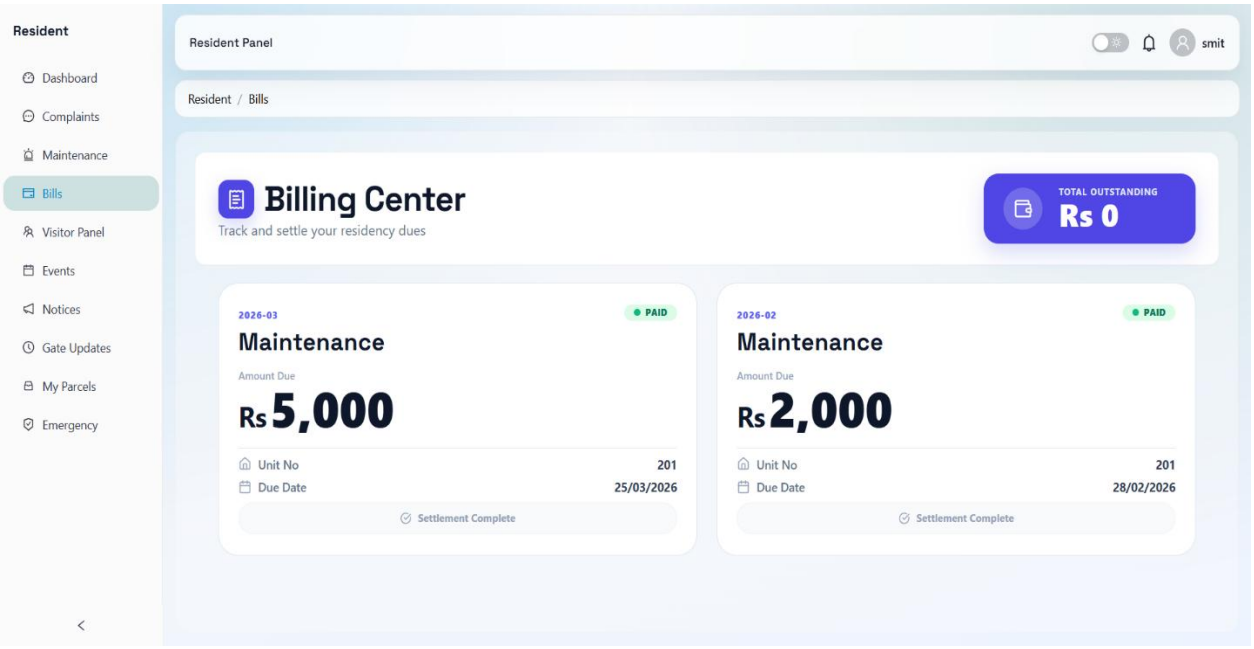
Complaints :-



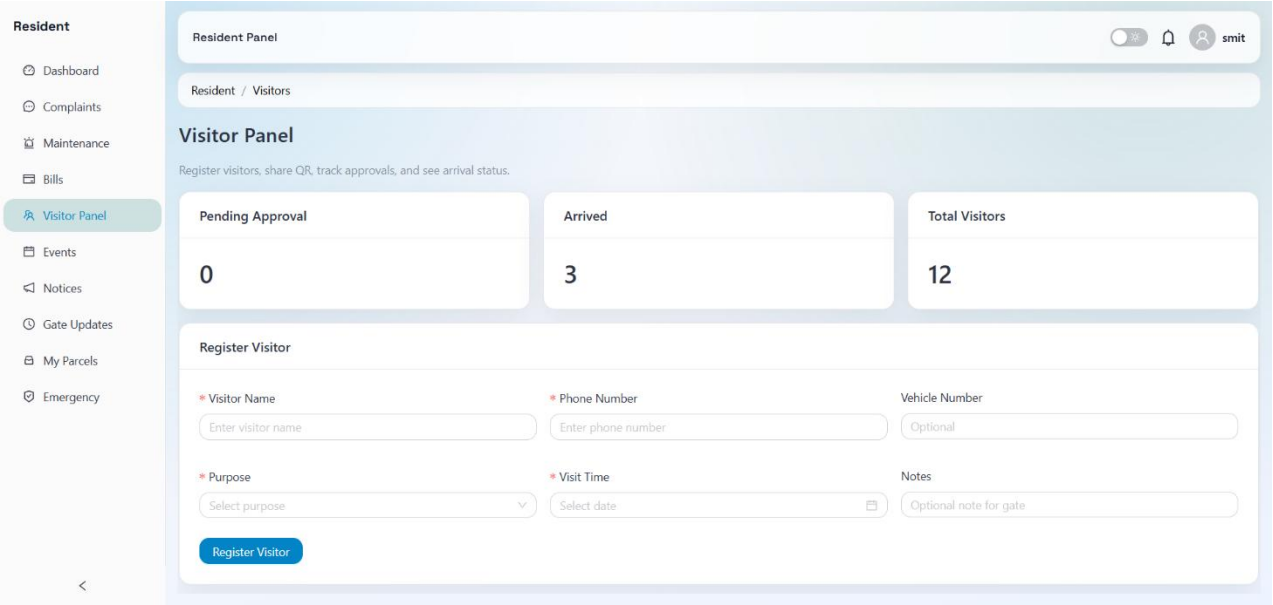
Maintenance:-



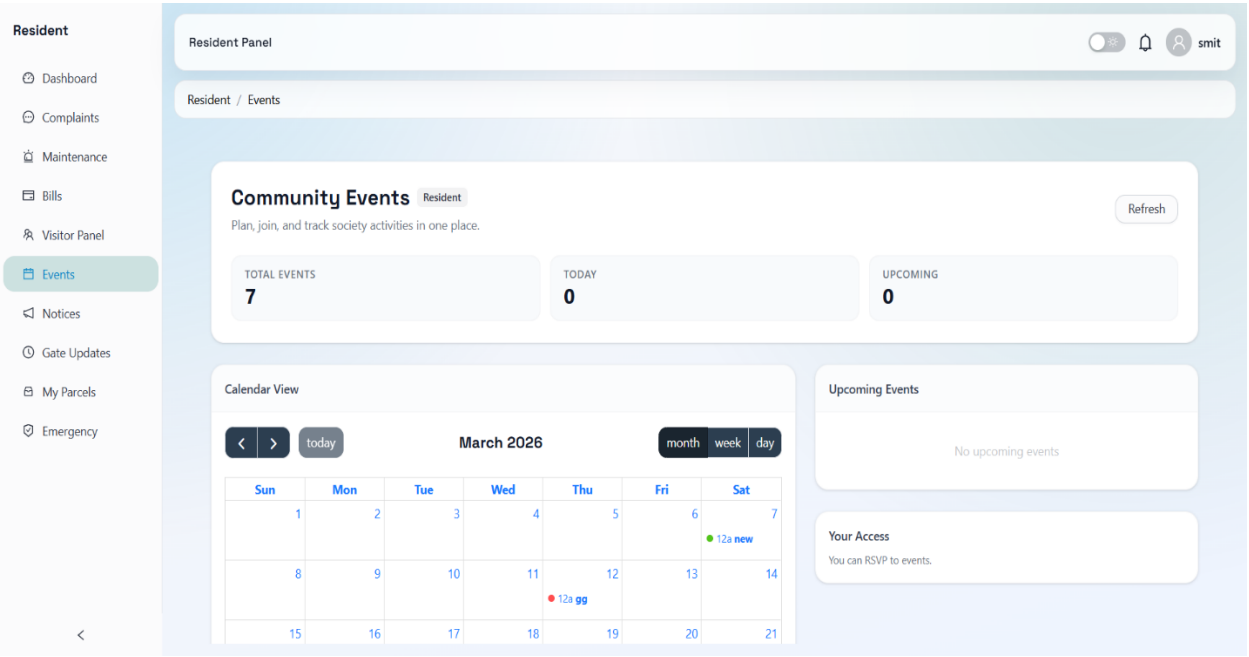
Bills:-



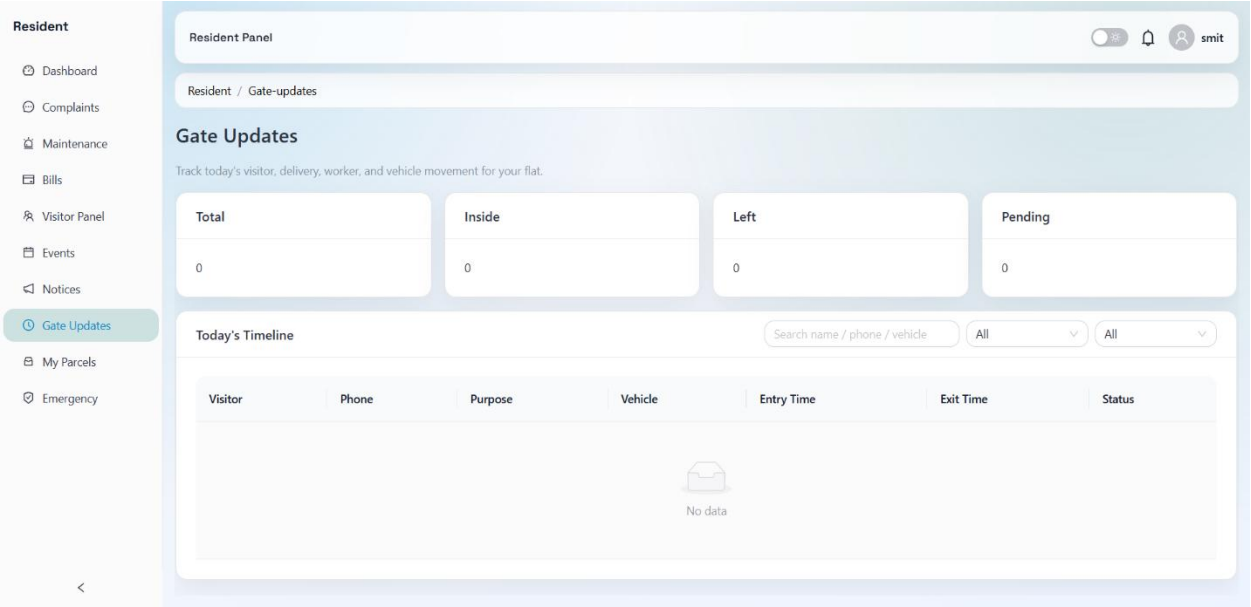
Visitor Panel :-



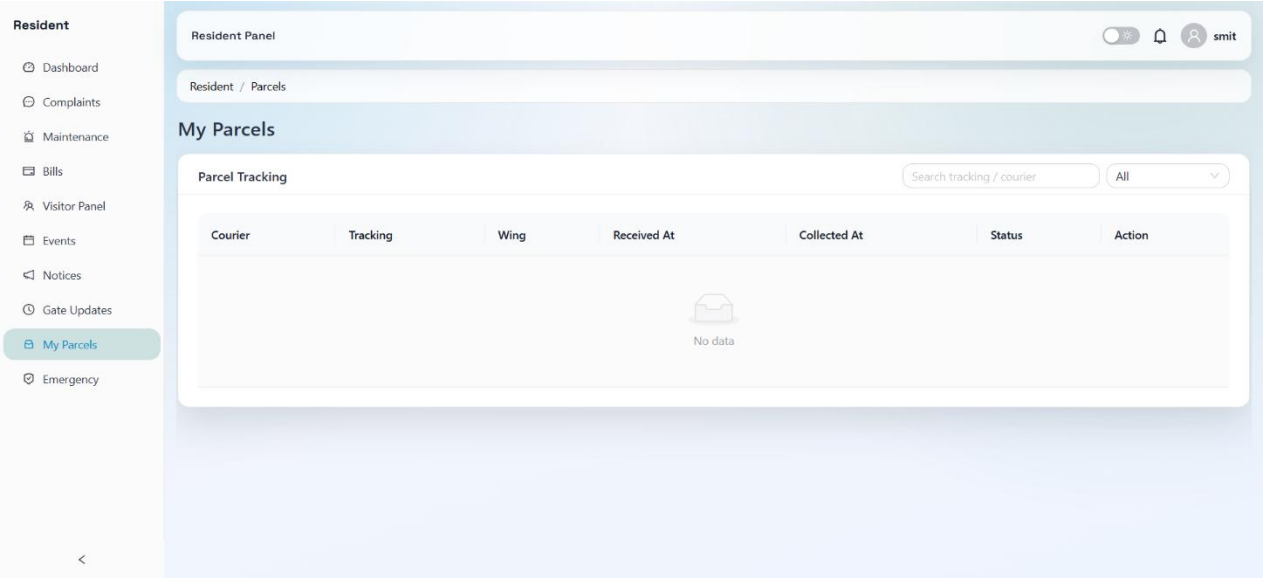
Events :-



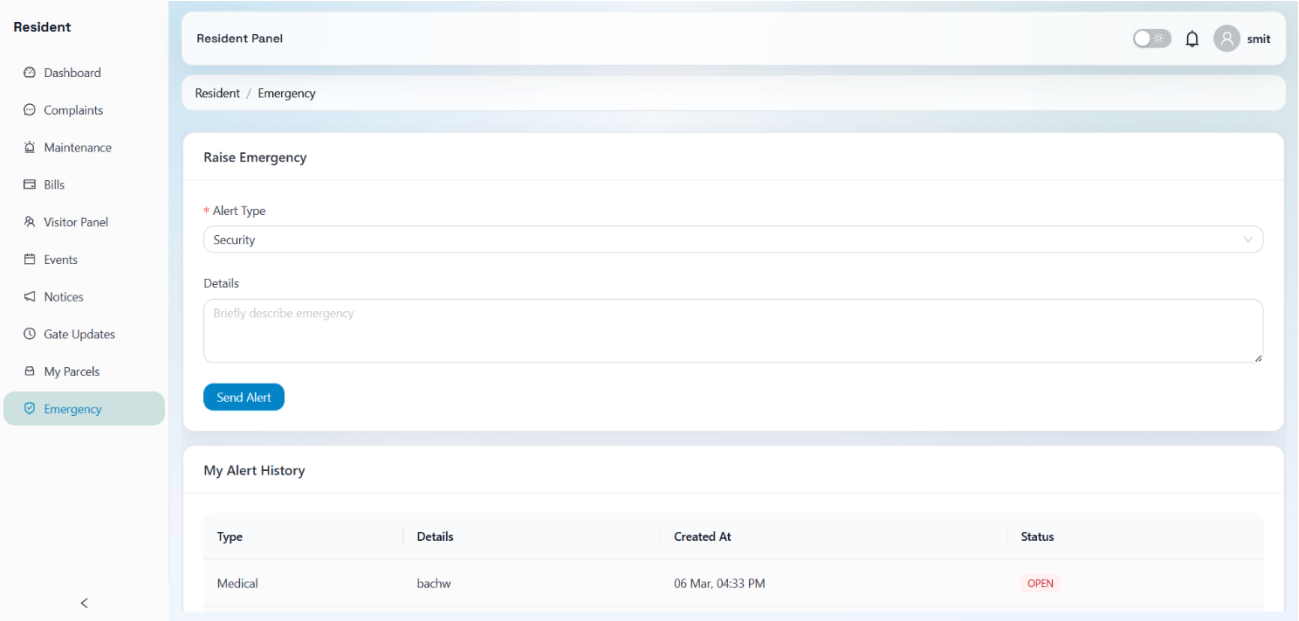
Gate Updates :-



My Parcels :-



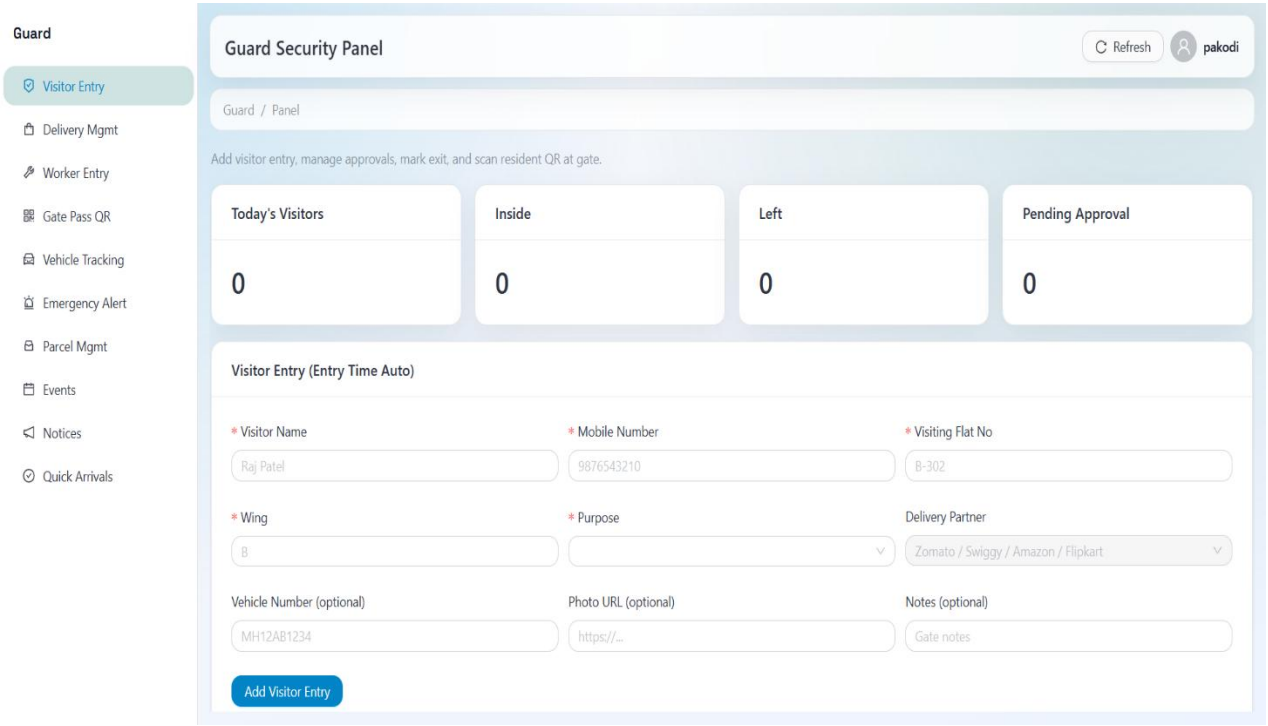
Emergency :-


Resident
Dashboard
Complaints
Maintenance
Bills
Visitor Panel
Events
Notices
Gate Updates
My Parcels
Emergency
Resident Panel
smit
Resident / Emergency
Raise Emergency
* Alert Type
Security
Details
Briefly describe emergency
Send Alert
My Alert History
Type
Details
Created At
Status
Medical
bachw
06 Mar, 04:33 PM
OPEN

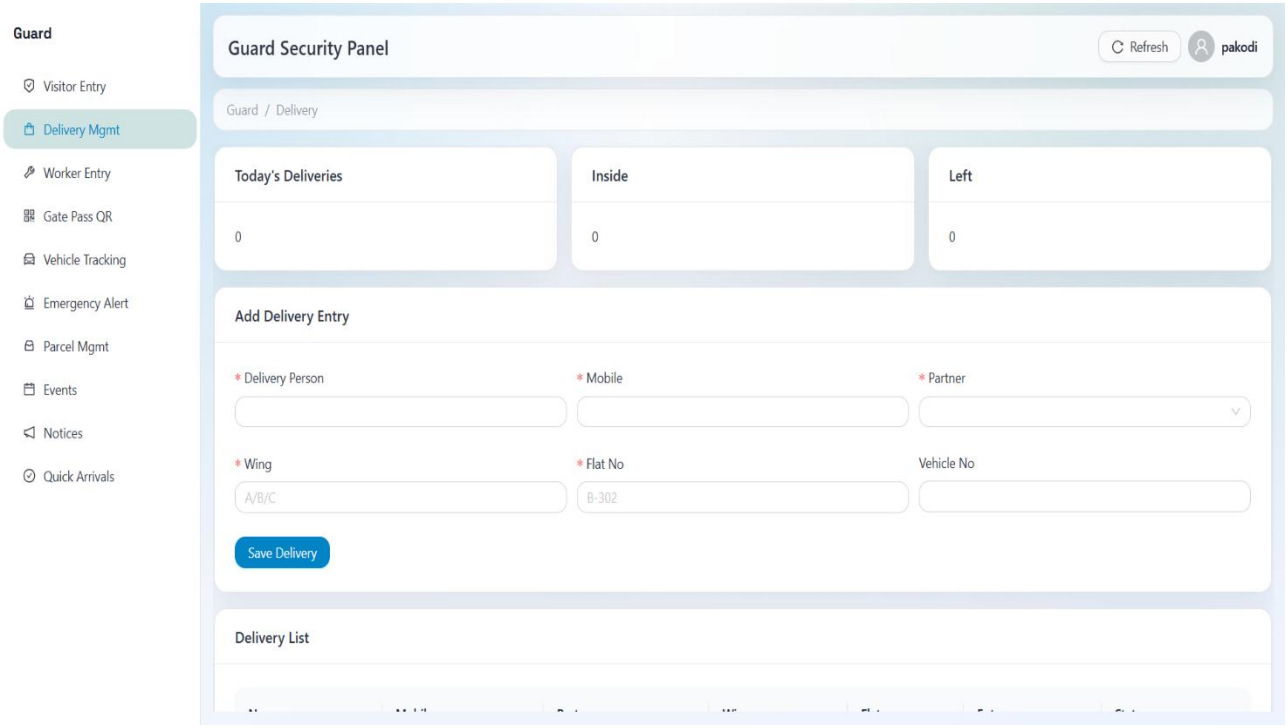
Style nest Clothing

Guard :-

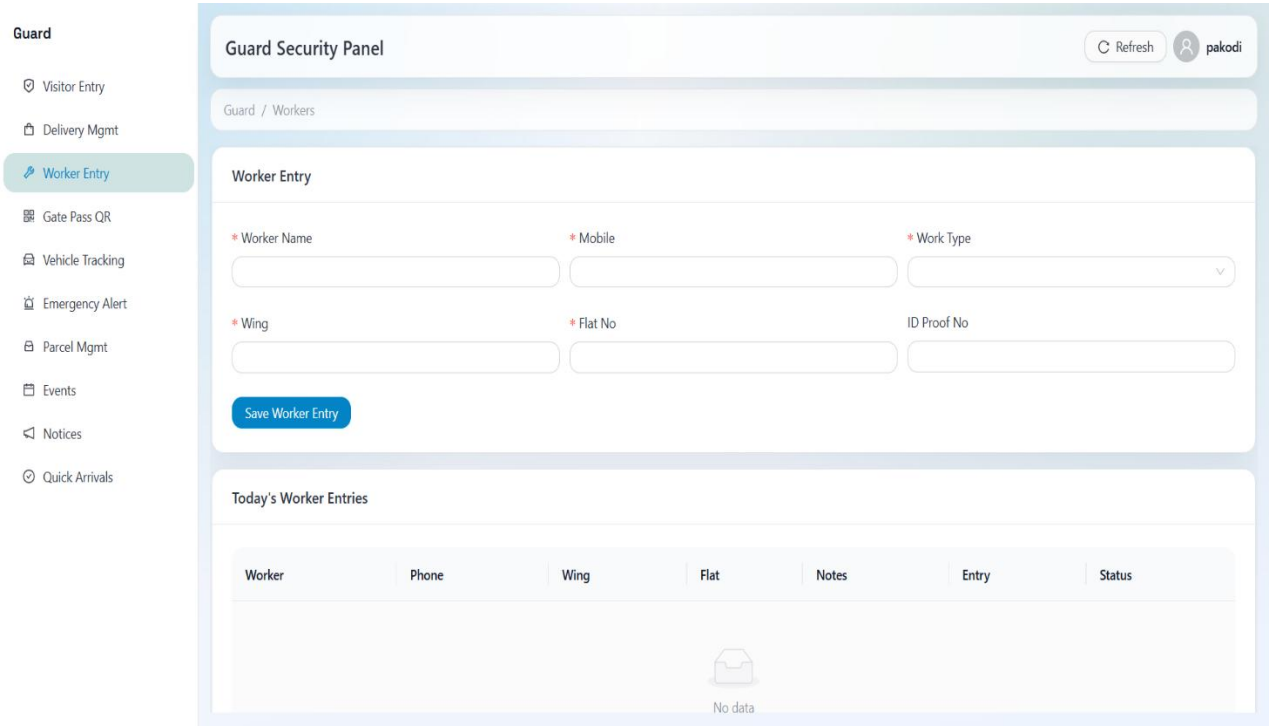
Dashboard :-



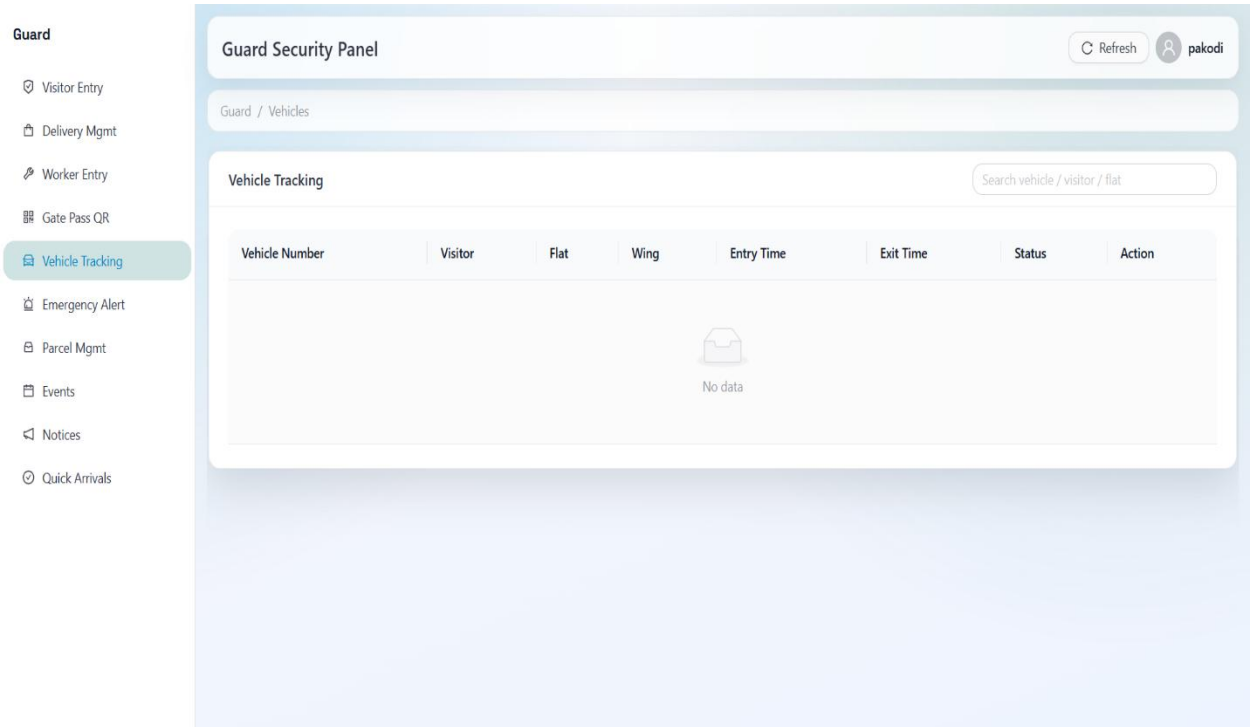
Delivery :-



Worker entry :-

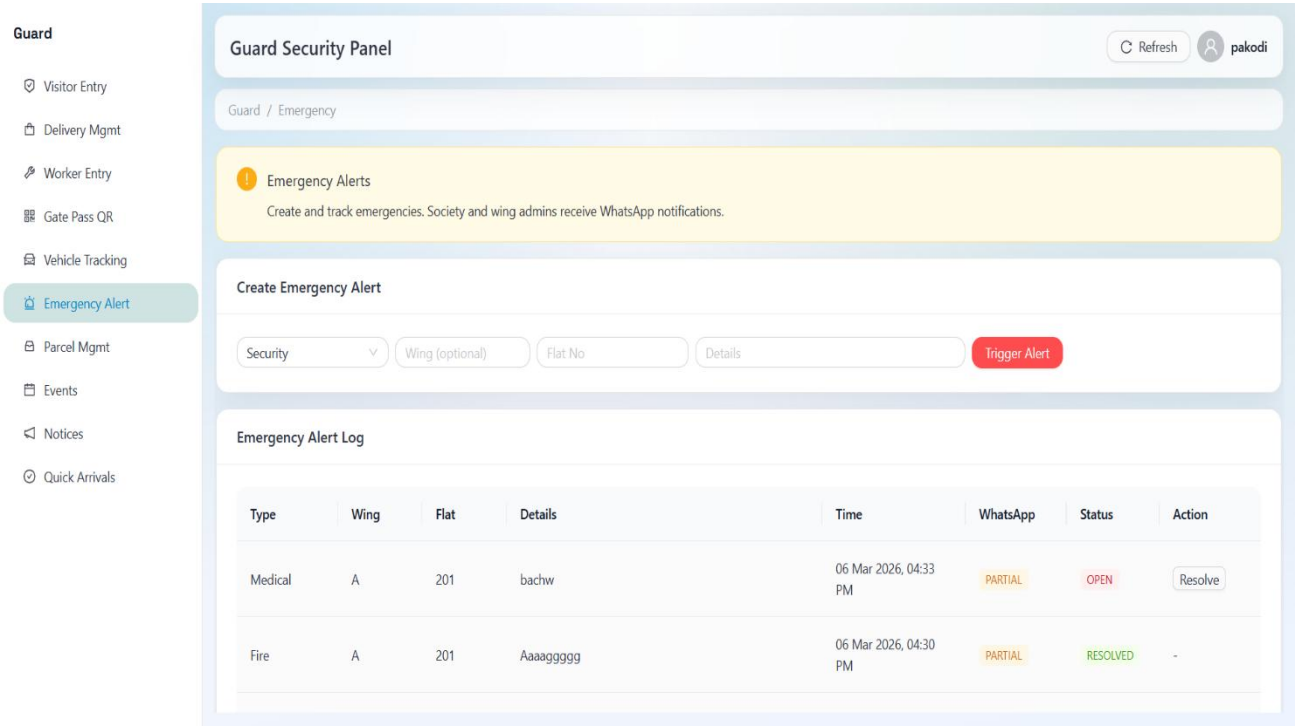


Vehicle Tracking :-

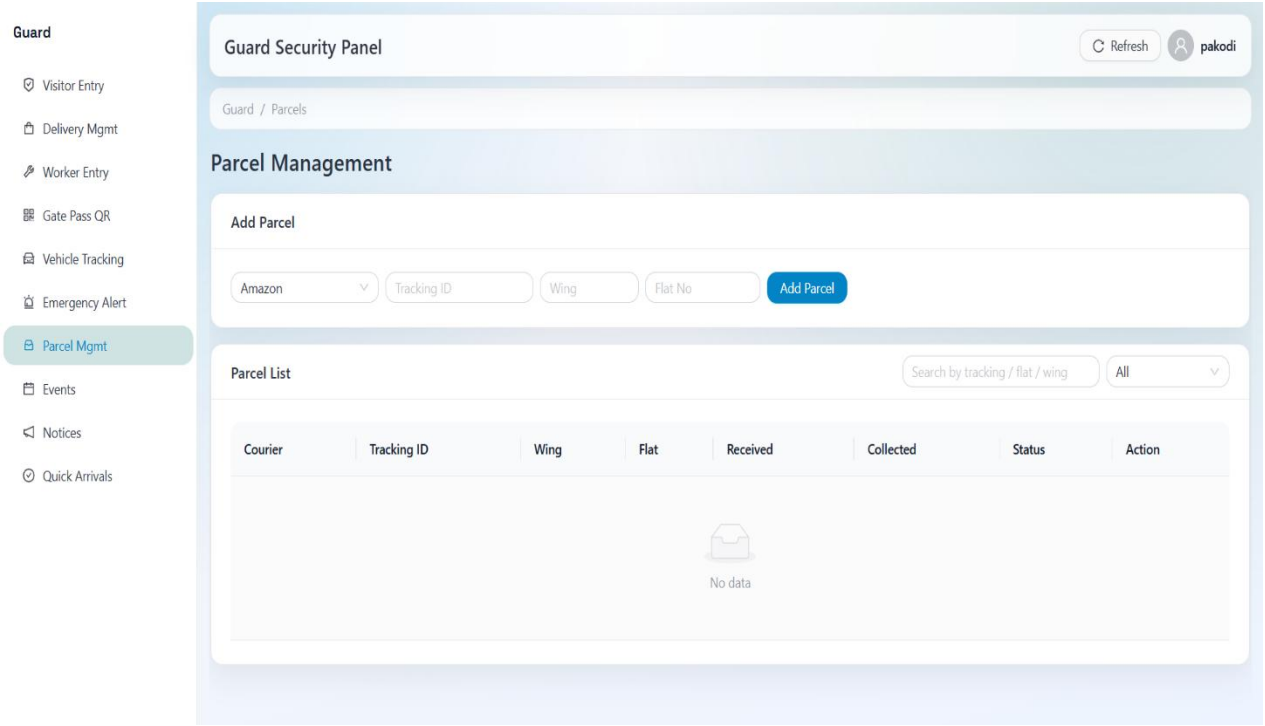


Style nest Clothing

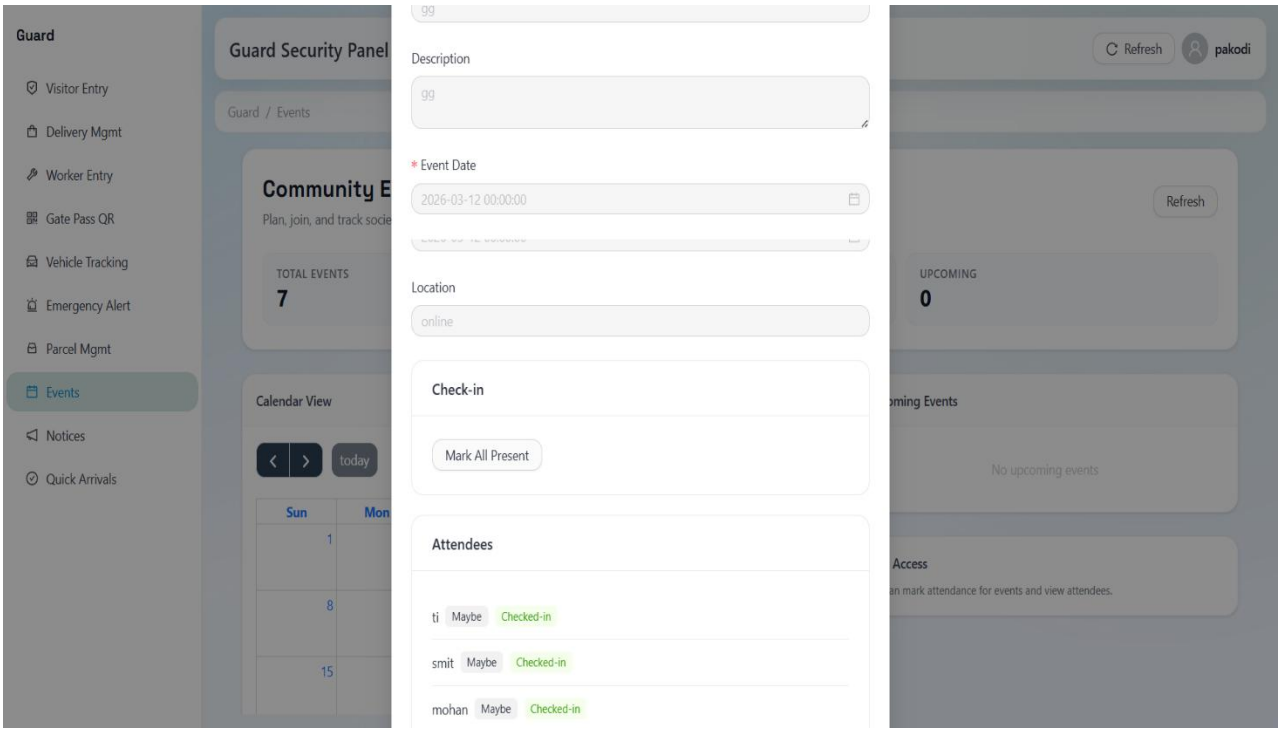
Emergency :-



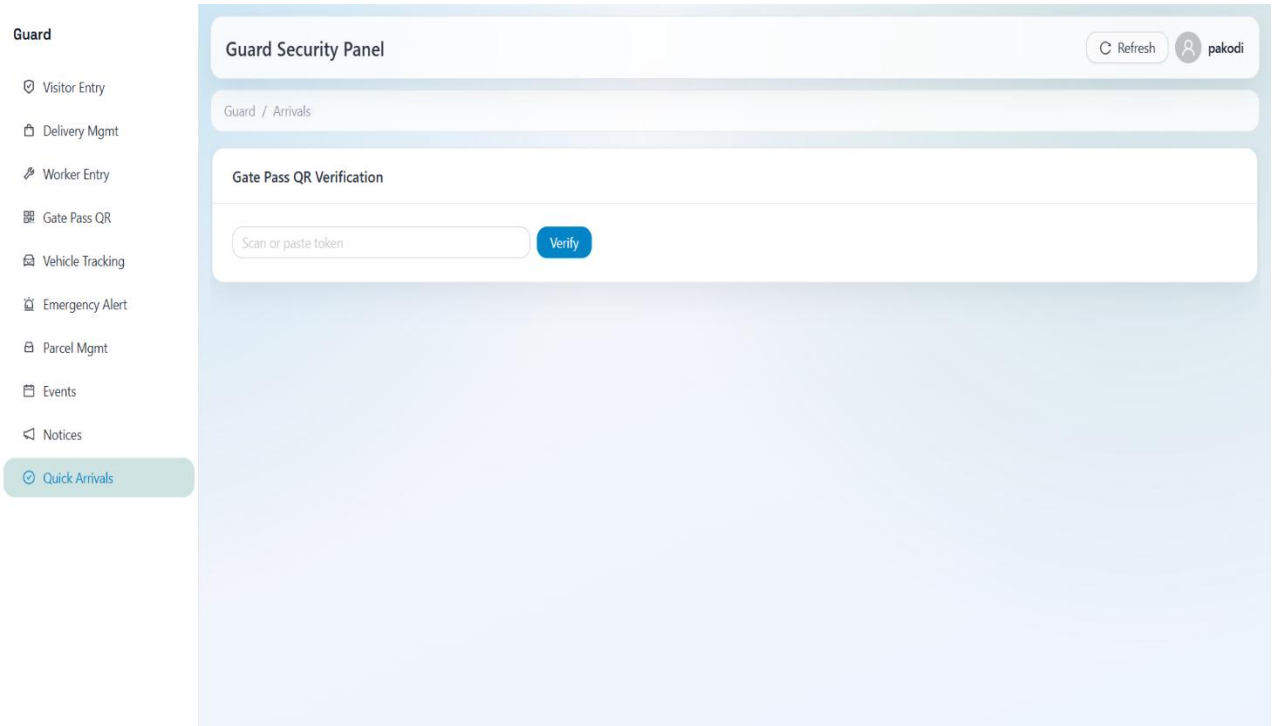
Parcel :-



Events :-

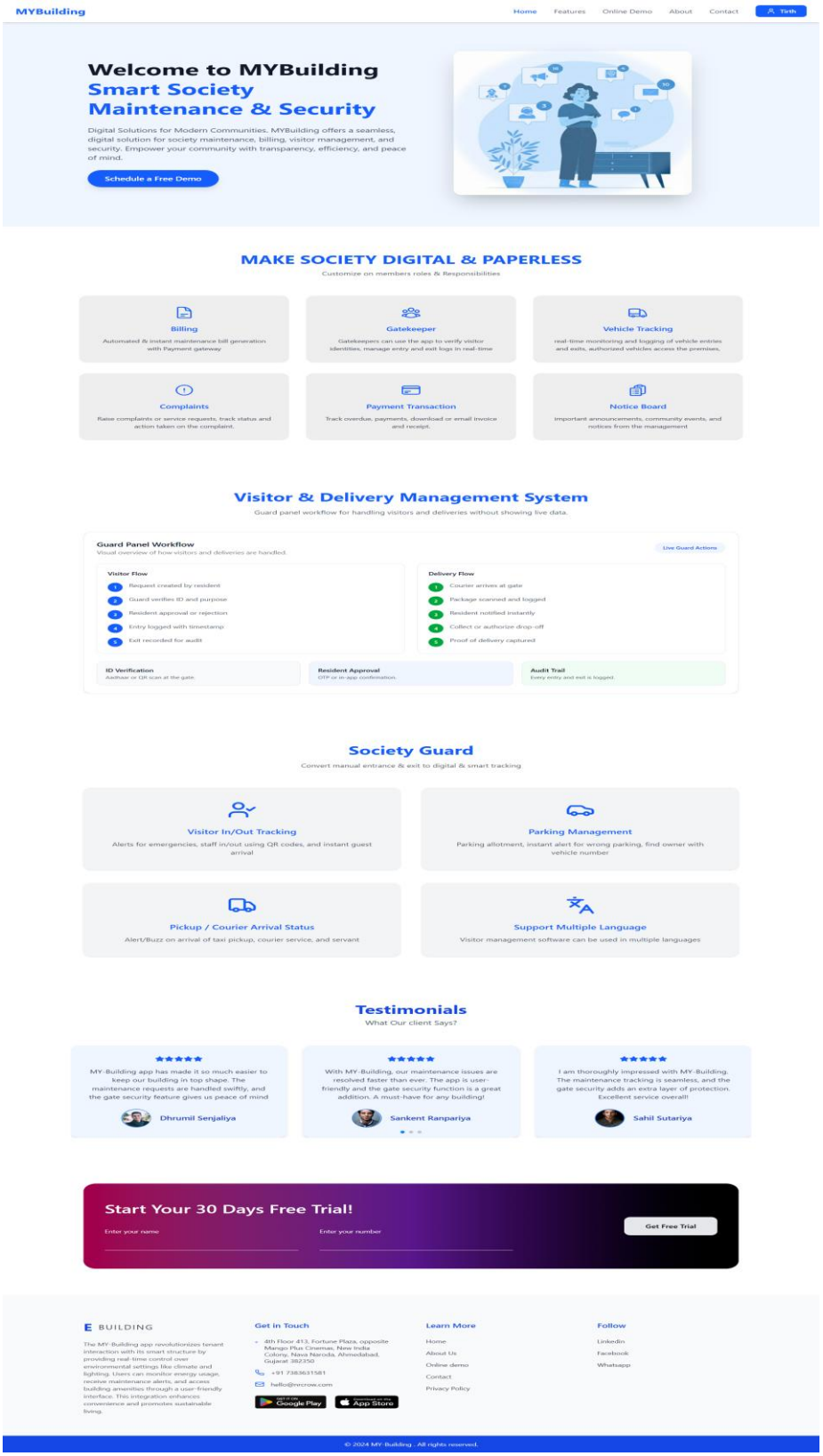


Quick Arrivals :-

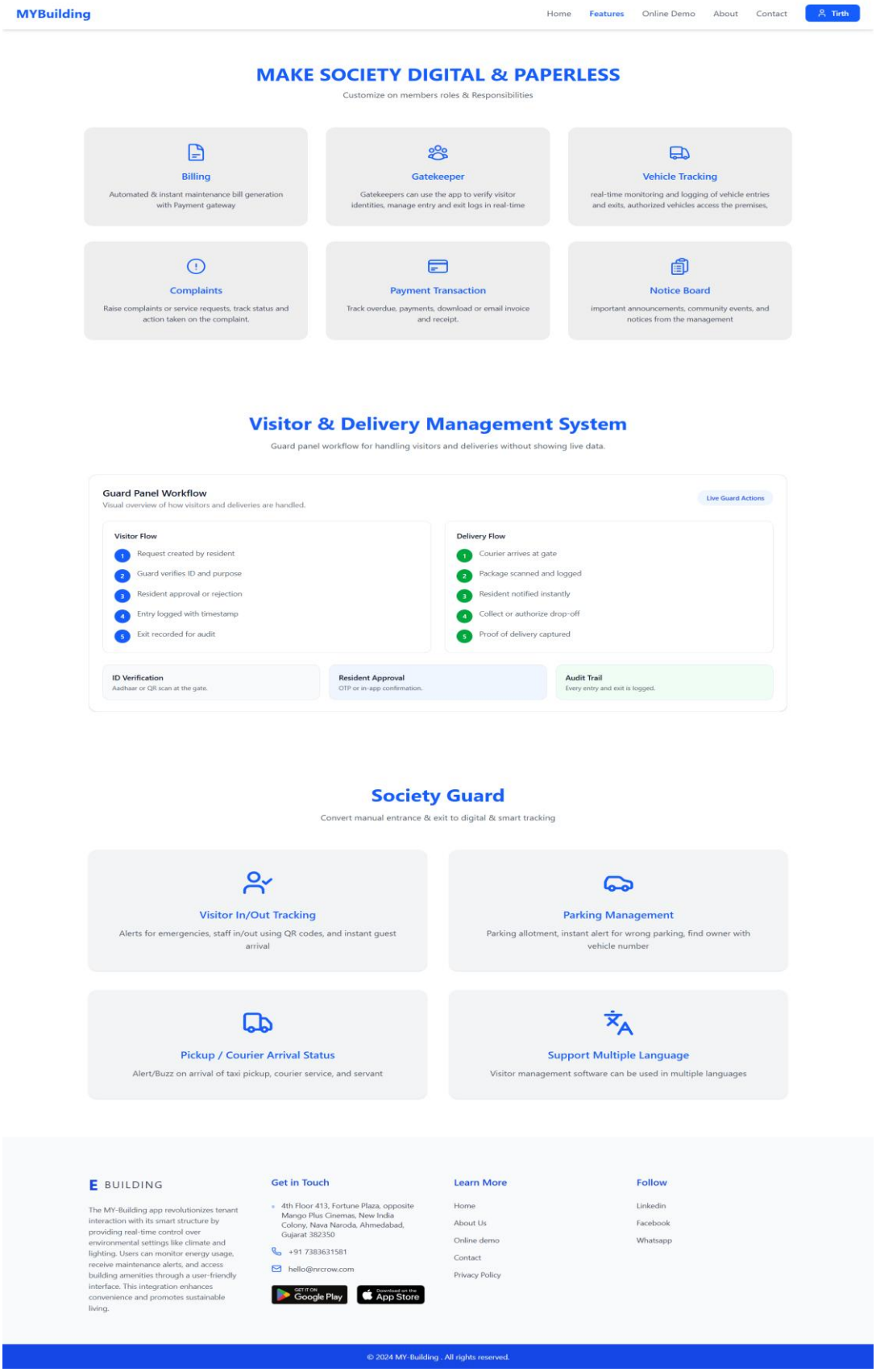


Demo Visitor :-

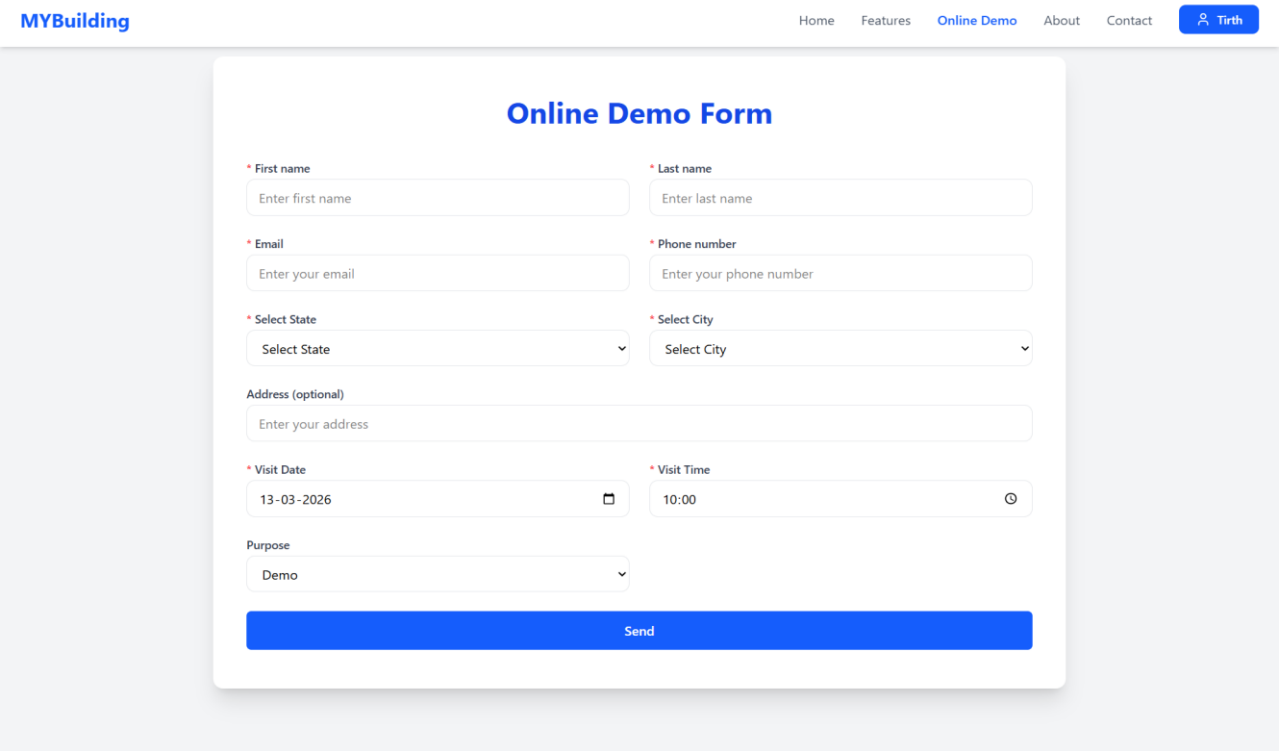
Home :-



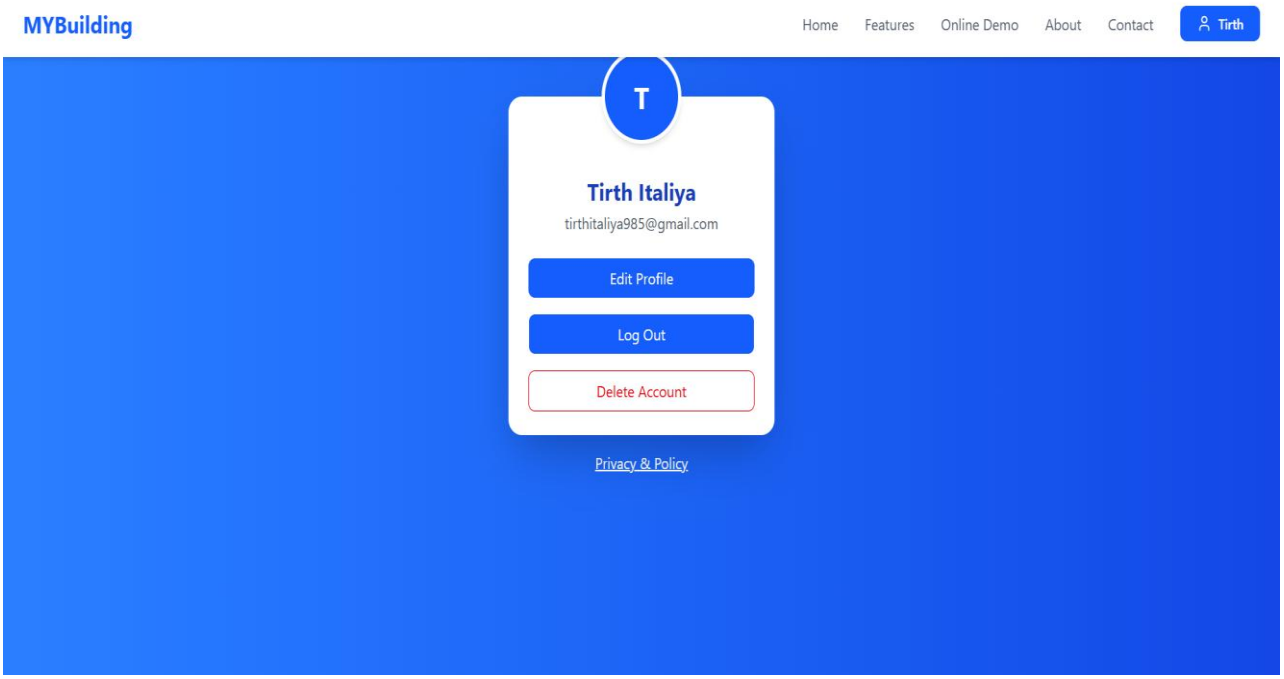
Features :-


MYBuilding
Home
Features
Online Demo
About
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MAKE SOCIETY DIGITAL & PAPERLESS
Customize on members roles & Responsibilities
Billing
Automated & instant maintenance bill generation with Payment gateway
Gatekeeper
Gatekeepers can use the app to verify visitor identities, manage entry and exit logs in real-time
Vehicle Tracking
real-time monitoring and logging of vehicle entries and exits, authorized vehicles access the premises,
Complaints
Raise complaints or service requests, track status and action taken on the complaint.
Payment Transaction
Track overdue, payments, download or email invoice and receipt.
Notice Board
important announcements, community events, and notices from the management
Visitor & Delivery Management System
Guard panel workflow for handling visitors and deliveries without showing live data.
Guard Panel Workflow
Visual overview of how visitors and deliveries are handled.
Live Guard Actions
Visitor Flow
1 Request created by resident
2 Guard verifies ID and purpose
3 Resident approval or rejection
4 Entry logged with timestamp
5 Exit recorded for audit
Delivery Flow
1 Courier arrives at gate
2 Package scanned and logged
3 Resident notified instantly
4 Collect or authorize drop-off
5 Proof of delivery captured
ID Verification
Aadhaar or QR scan at the gate.
Resident Approval
OTP or in-app confirmation.
Audit Trail
Every entry and exit is logged.
Society Guard
Convert manual entrance & exit to digital & smart tracking
Visitor In/Out Tracking
Alerts for emergencies, staff in/out using QR codes, and instant guest arrival
Parking Management
Parking allotment, instant alert for wrong parking, find owner with vehicle number
Pickup / Courier Arrival Status
Alert/Buzz on arrival of taxi pickup, courier service, and servant
Support Multiple Language
Visitor management software can be used in multiple languages
E BUILDING
The MY-Building app revolutionizes tenant interaction with its smart structure by providing real-time control over environmental settings like climate and lighting. Users can monitor energy usage, receive maintenance alerts, and access building amenities through a user-friendly interface. This integration enhances convenience and promotes sustainable living.
Get in Touch
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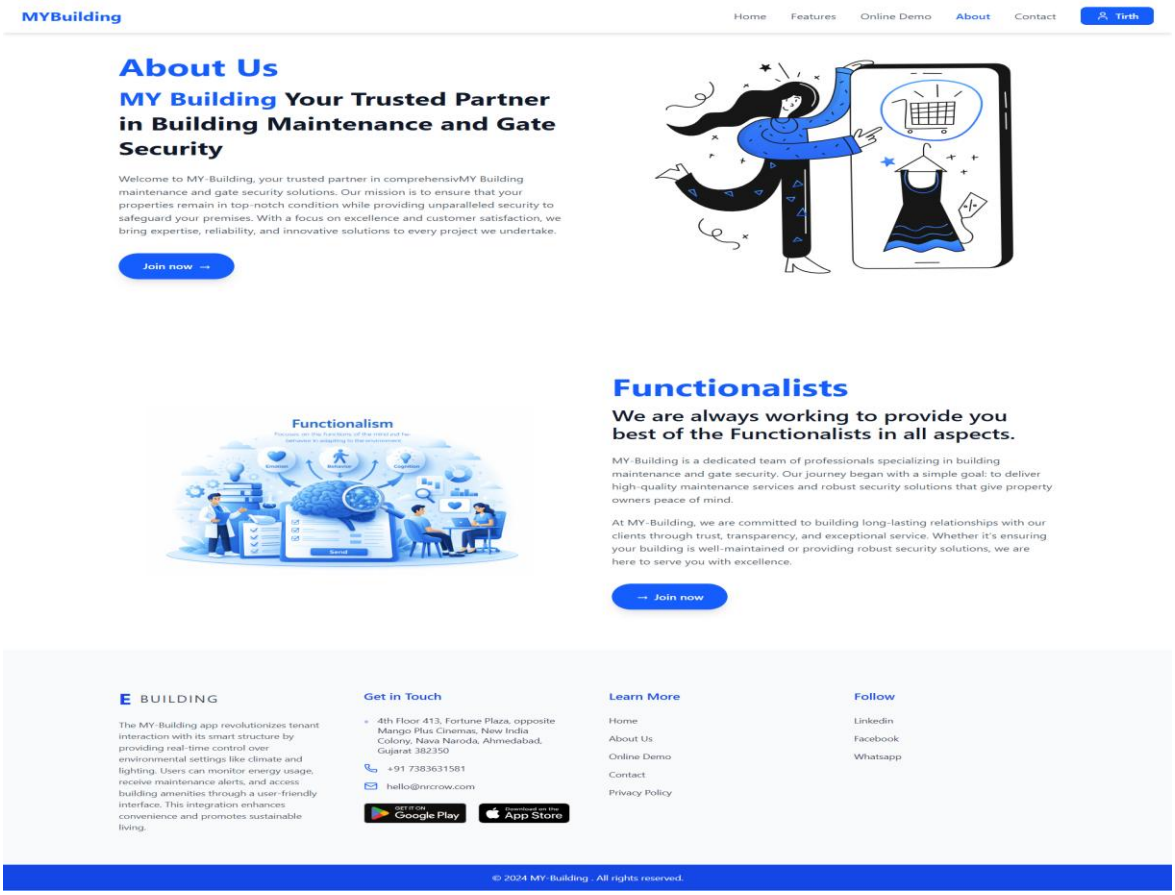
Online Demo:-



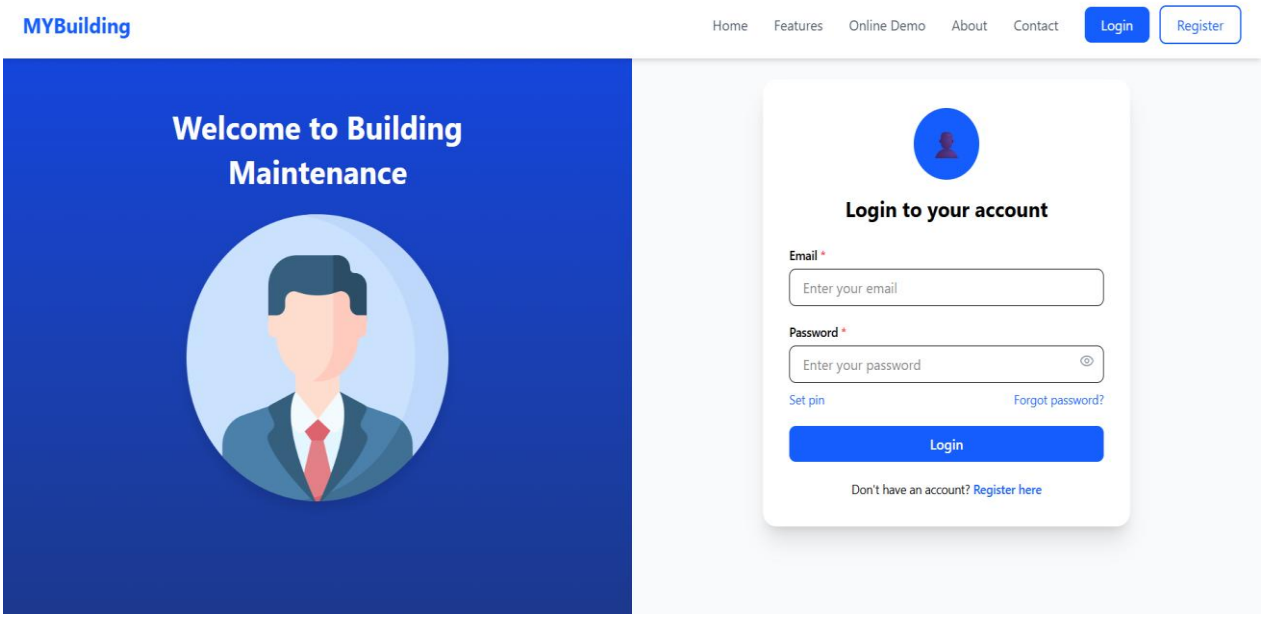
Profile :-



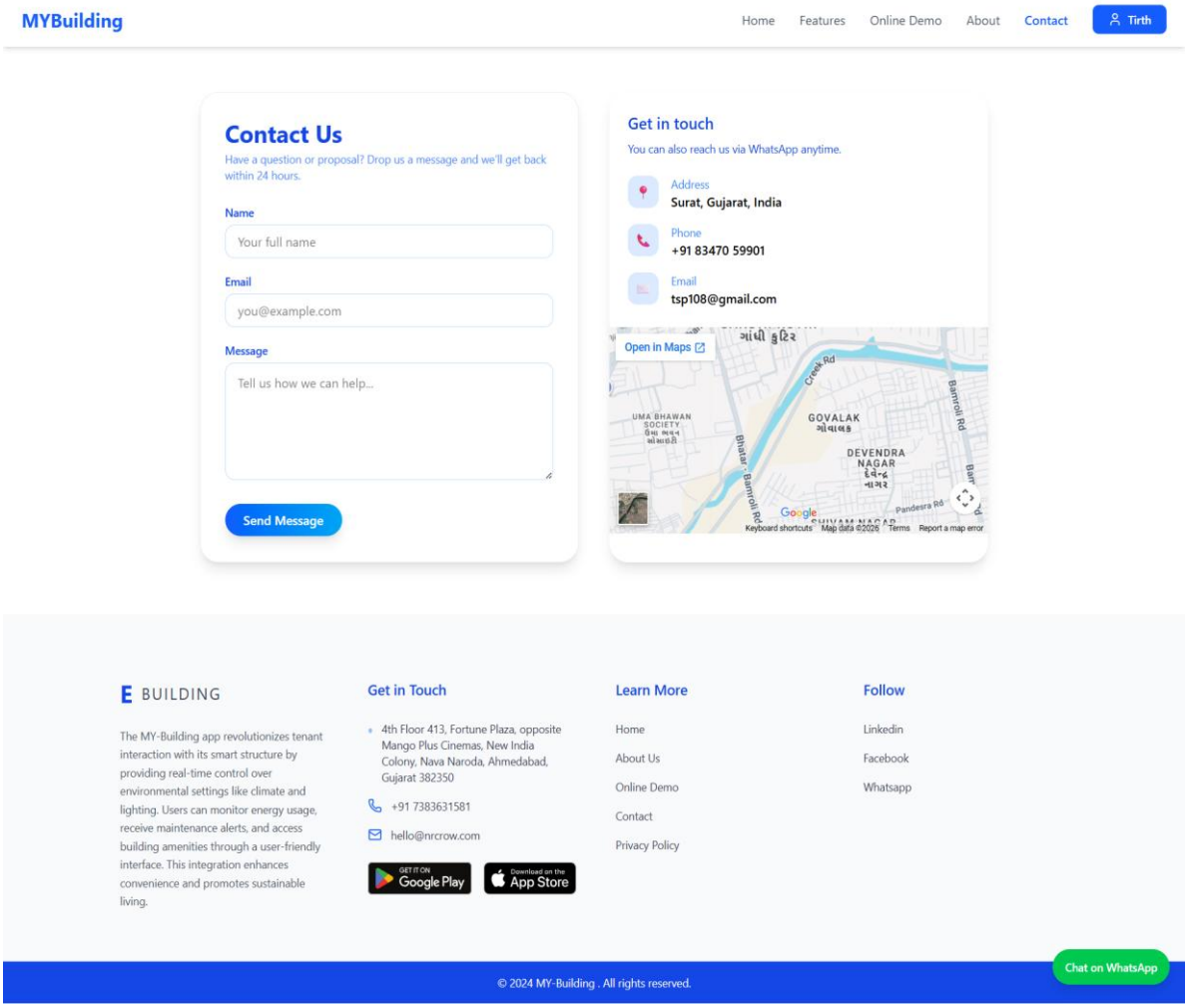
About :-


MYBuilding
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About Us
MY Building Your Trusted Partner in Building Maintenance and Gate Security
Welcome to MY-Building, your trusted partner in comprehensivMY Building maintenance and gate security solutions. Our mission is to ensure that your properties remain in top-notch condition while providing unparalleled security to safeguard your premises. With a focus on excellence and customer satisfaction, we bring expertise, reliability, and innovative solutions to every project we undertake.
Join now →
Functionalism
Functionalists
We are always working to provide you best of the Functionalists in all aspects.
MY-Building is a dedicated team of professionals specializing in building maintenance and gate security. Our journey began with a simple goal: to deliver high-quality maintenance services and robust security solutions that give property owners peace of mind.
At MY-Building, we are committed to building long-lasting relationships with our clients through trust, transparency, and exceptional service. Whether it's ensuring your building is well-maintained or providing robust security solutions, we are here to serve you with excellence.
→ Join now
E BUILDING
The MY-Building app revolutionizes tenant interaction with its smart structure by providing real-time control over environmental settings like climate and lighting. Users can monitor energy usage, receive maintenance alerts, and access building amenities through a user-friendly interface. This integration enhances convenience and promotes sustainable living.
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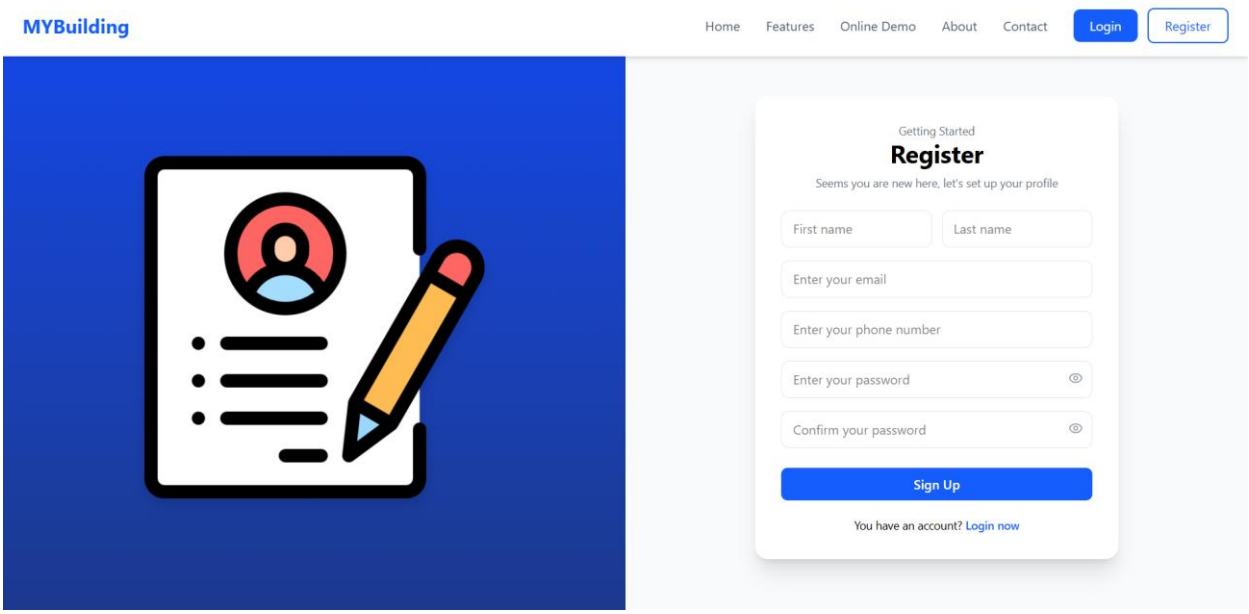
Login :-


MYBuilding
Home
Features
Online Demo
About
Contact
Login
Register
Welcome to Building Maintenance
Login to your account
Email *
Enter your email
Password *
Enter your password
Set pin
Forgot password?
Login
Don't have an account? Register here

Contact :-



Register :-



Conclusion :-

- The Society Management System was developed to simplify and digitize the management of residential societies. This system reduces manual work and improves communication between administrators, residents, and security staff.
- The project provides role-based access for different users such as Admin, Society Admin, Wing Admin, Resident, Guard, and Demo Visitor, allowing each user to perform tasks according to their responsibilities. Features such as maintenance management, visitor management, bill generation, payment integration, and profile management help automate daily society operations.
- By implementing this system, society management becomes more efficient, transparent, and secure. Residents can easily access services, pay maintenance online, and receive updates, while administrators can monitor and manage society activities effectively.
- Overall, the Society Management System demonstrates how modern web technologies and database management can be used to create a practical solution for real-world residential community management.

Websites:-

- HTML Tutorial :- https://www.geeksforgeeks.org/html/html-tutorial/
- CSS Documentation :- https://www.w3schools.com/css/
- JavaScript Doc. :- https://www.w3schools.com/js/
- ReactJS Tutorial :- https://www.w3schools.com/REACT/DEFAULT.ASP
- NodeJS Documentation :- https://nodejs.org/en
- ExpressJS Tutorial :- https://www.geeksforgeeks.org/node-js/express-js/
- MongoDB Database :- https://github.com/mongodb/mongo
- System Flow chart & Diagrames :- https://creately.com/diagram-community/popular
- All basic Details :- https://www.wikipedia.org/
- Taking all website' Screenshots :- https://gofullpage.com/

These resources provided detailed explanations, code samples, and best practices that helped in understanding the technologies and implementing the project successfully.